# Hallucination in Large Language Models: From Mechanistic Understanding to Novel Control Frameworks

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2024 7th International Conference on Universal Village (UV) | 979-8-3315-3151-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/UV63228.2024.11189226

***Abstract*—Hallucination in large language models (LLMs) represents a fundamental challenge that extends beyond technical failure to reveal the mathematical inevitability of uncertainty in probabilistic language generation. This comprehensive survey presents the first unified mechanistic framework that systematically maps causal pathways from internal mechanisms to observable manifestations. We introduce a hierarchical taxonomy organizing hallucinations into four manifestation categories (P1: factuality errors, P2: faithfulness violations, P3: logical inconsistencies, P4: emergent behaviors) and trace their origins to three mechanistic sources (C1: data artifacts, C2: training biases, C3: inference failures). Our analysis reveals the dual nature of hallucination, simultaneously representing creative potential and factual unreliability, which challenges assumptions about complete elimination. We present seven novel frameworks for mechanism-aware hallucination management: HALO-M for layer-wise diagnostic analysis, KOQF for quantifying knowledge overshadowing effects, MIT for targeted mechanistic interventions, DPHCS for dual-process adaptive generation, HABS for comprehensive benchmarking, NSHPA for neuro-symbolic verification, and HACHMS for human-AI collaborative management. Through systematic task decomposition and compositional verification, these frameworks enable precise identification and mitigation of hallucination at its source. By establishing hallucination as an emergent property to be understood and managed rather than suppressed, we provide both theoretical foundations and practical tools for developing AI systems that balance creative capability with factual reliability, ultimately reframing the hallucination challenge from an obstacle to an opportunity for advancing artificial intelligence aligned with human values.**

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***Keywords*—hallucination, large language models, mechanistic interpretability, causal pathways, taxonomy, faithfulness and factuality, human factors, reliability and safety, inference and decoding, data artifacts and optimization bias.**

## I. Introduction

Large language models (LLMs) have transformed from research curiosities to critical infrastructure powering healthcare diagnostics [1], [2], legal analysis [3], [4], educational systems [5], and creative industries [6], [7]. Yet beneath their remarkable capabilities lies a fundamental paradox: the same mechanisms that enable creative synthesis and flexible reasoning also produce hallucinations, which are confident, fluent outputs that appear authoritative but lack grounding in reality. As these systems become increasingly embedded in high-stakes applications, understanding and managing hallucination has evolved from an academic concern to an urgent societal imperative [8], [9].

Recent theoretical advances have revealed a startling insight: hallucination is not merely a technical flaw to be debugged but a mathematical inevitability arising from the probabilistic nature of language generation itself [10], [11]. This discovery fundamentally challenges our approach to the problem. Rather than pursuing the quixotic goal of eliminating hallucination entirely, we must develop sophisticated frameworks to understand, predict, and manage these phenomena based on their underlying mechanisms. This shift parallels historical transitions

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in other fields, from attempting to eliminate uncertainty in quantum mechanics to embracing it as a fundamental property that enables quantum computation.

Existing surveys have made valuable contributions by cataloging hallucination types, documenting mitigation strategies, and proposing evaluation metrics [12]–[14]. However, they primarily describe what we observe rather than explaining why it happens. This descriptive approach, while useful for practitioners, fails to provide the mechanistic understanding necessary for developing targeted interventions. The gap between phenomenology and mechanism represents a critical barrier to progress, leaving researchers and practitioners to apply broad brush solutions to problems they do not fully understand [15].

This survey bridges that gap by presenting the first comprehensive mechanistic framework that traces hallucination from its computational origins to its observable manifestations. We move beyond surface-level categorization to reveal the causal pathways that connect data artifacts, training dynamics, architectural constraints, and inference procedures to specific types of hallucination. Our analysis reveals the dual nature of hallucination as both a source of creative potential and factual unreliability, demonstrating how the same mechanisms that produce harmful fabrications also enable beneficial creative synthesis [16], [17].

### *A. Research Questions and Approach*

Our investigation is guided by four fundamental research questions that progressively deepen our understanding:

- **RQ1**: What are the precise computational mechanisms that give rise to different classes of hallucination, and how can we trace specific manifestations back to their mechanistic origins?
- **RQ2**: How do factors across the model lifecycle, from training data to architectural choices to inference procedures, interact to create distinct hallucination pathways?
- **RQ3**: What role do human factors play in both causing and mitigating hallucination, and how can we design human-AI systems that leverage complementary strengths?
- **RQ4**: How can mechanistic understanding inform the development of targeted interventions that address root causes rather than surface symptoms?

To answer these questions, we adopt a multilayered analytical approach that combines mechanistic interpretability, causal analysis, and systems thinking. We systematically decompose the hallucination phenomenon into its constituent parts, examine their interactions, and reconstruct a comprehensive understanding that spans from microscopic neural activations to macroscopic system behaviors [18], [19].

### *B. Key Contributions*

This survey makes seven major contributions that advance both theoretical understanding and practical application:

1) **Unified Mechanistic Framework**: We present the first systematic mapping between internal mechanisms (C1: data-level, C2: training-induced, C3: inference-stage) and external manifestations (P1: factuality errors, P2: faithfulness violations, P3: logical inconsistencies, P4: emergent behaviors), revealing precise causal pathways that enable targeted interventions [15], [20].
2) **Comprehensive Taxonomy with Mechanism Grounding**: Our hierarchical taxonomy goes beyond surface categorization to connect each manifestation type with its underlying computational mechanisms, providing actionable insights for detection and mitigation.
3) **Seven Novel Management Frameworks**: We introduce:
   - HALO-M: Layer-wise hallucination analysis for real-time mechanism detection [21]
   - KOQF: Mathematical framework for quantifying knowledge overshadowing effects [22]
   - MIT: Modular toolkit for mechanism-specific interventions
   - DPHCS: Dual-process system implementing fast/slow thinking modes
   - HABS: Comprehensive benchmark suite organized by mechanism-manifestation pathways
   - NSHPA: Neuro-symbolic architecture combining neural flexibility with symbolic verification
   - HACHMS: Human-AI collaborative system for optimal task distribution
4) **Task Decomposition and Compositional Verification**: We demonstrate how decomposing complex queries into atomic units enables precise hallucination localization and targeted mitigation, transforming opaque generation into interpretable processes [23], [24].
5) **Human Factors as Primary Considerations**: We provide the most comprehensive analysis to date of how human decisions, biases, and interactions shape hallucination throughout the AI lifecycle, from training data annotation to end user interactions.
6) **Paradigm Shift from Elimination to Management**: We establish hallucination as an emergent property to be understood and managed rather than eliminated, providing frameworks for navigating the trade-offs between creative capability and factual reliability [10], [16], [17].
7) **Actionable Guidance for Practitioners**: We translate mechanistic insights into concrete recommendations for model developers, system designers, and end users, bridging the gap between theory and practice.

### *C. Survey Organization*

This survey is structured to progressively build understanding from observable phenomena to underlying mechanisms to future solutions:

Section II situates our work within the rapidly evolving landscape of hallucination research, highlighting gaps in existing approaches and establishing the need for mechanistic understanding. Section III presents our comprehensive taxonomy of hallucination manifestations, organizing observable patterns into a hierarchical framework that facilitates systematic analysis. Section IV examines factors across the model lifecycle

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that contribute to hallucination, establishing the causal context for mechanistic analysis.

Section V forms the theoretical core of our survey, revealing the internal computational mechanisms that produce hallucination and mapping them to specific manifestations. Section VI explores the critical yet often overlooked role of human factors, demonstrating how human decisions and biases shape hallucination throughout the AI lifecycle. Section VII presents our seven novel frameworks for mechanism-aware hallucination management, offering concrete tools for detection, prediction, and mitigation. Finally, Section VIII synthesizes insights across all sections, discussing implications for AI development and outlining critical directions for future research.

**Notation and labels.** Mechanism categories: C1 (from data), C2 (from training), C3 (from inference). Submechanisms are denoted Ck–Ti (e.g., C1–T3 = knowledge limitations). Manifestation categories: P1 (factuality), P2 (faithfulness), P3 (logical), P4 (presentation/emergent). Manifestations follow Pj–Lk–Ti format where L denotes level and T denotes type. Where helpful we use links such as Ck–Ti $\rightarrow$ Pj–L$\ell$. Section references: mechanisms (Section V), manifestations (Section III), contributing factors (Section IV).

## II. Background and Related Work

We position our survey within four strands of rapidly evolving work (2023–2025): comprehensive surveys, mechanistic interpretability, multimodal hallucination, and evaluation benchmarks.

### A. Surveys of Hallucination in LLMs

Recent surveys provide broad overviews of definitions, taxonomies, and mitigations, but primarily describe phenomena rather than mechanisms [8], [9], [13], [25]. Our work complements these by connecting observable patterns to causal pathways, bridging phenomenology with mechanistic accounts.

### B. Mechanistic Interpretability and Causal Accounts

Mechanistic interpretability has begun to illuminate how transformers compute, offering tools to trace circuits, features, and optimization dynamics [18], [19]. We build on these insights to hypothesize and organize mechanisms that plausibly underlie distinct hallucination classes, emphasizing how data artifacts and optimization biases interact with architectural constraints.

### C. Multimodal Hallucination

Hallucination extends beyond text-only models. Object hallucination in image captioning and multimodal generation reveals modality-specific and cross-modal drivers [26], [27]. These findings inform our taxonomy by highlighting mechanisms shared across modalities (e.g., prior over-reliance, weak grounding) and those unique to alignment and fusion.

### D. Evaluation and Benchmarks

Evaluation ranges from task-specific probes to broader audit perspectives and leaderboards. While metrics vary, a common limitation is entangling surface symptoms with causes. Efforts like HALO analyze hallucination patterns more structurally [28], and recent surveys catalog benchmark coverage and gaps [9]. We advocate mechanism-aligned evaluation that tests causal hypotheses and intervention-target fit.

## III. Manifestations of Hallucination in Large Language Models

Hallucination in large language models manifests as a complex spectrum of erroneous outputs that systematically deviate from factual accuracy, contextual faithfulness, or logical consistency. Unlike traditional rule-based systems where errors follow predictable, deterministic patterns, LLM hallucinations emerge from the intricate interplay between learned statistical associations, attention mechanisms, and autoregressive generation processes, creating a rich taxonomy of failure modes that reflect fundamental challenges in neural language understanding and generation.

The systematic study of hallucination manifestations serves dual purposes: it provides the empirical foundation for developing targeted detection and mitigation strategies, while simultaneously offering crucial insights into the computational limitations and failure modes of current neural architectures. Understanding these external manifestations is essential for bridging the gap between observable model behavior and the underlying mechanisms explored in Section V, as each manifestation type provides valuable clues about the specific computational processes that give rise to erroneous outputs.

This section presents a comprehensive taxonomy of hallucination manifestations organized along three primary dimensions—factuality, faithfulness, and logical consistency—that collectively capture the full spectrum of observable errors in contemporary LLMs and multimodal systems. Our taxonomic approach moves beyond simple error cataloging to establish a systematic framework that reveals the relationships between different manifestation types, their underlying computational origins, and their implications for system reliability. Each manifestation category represents distinct patterns of deviation from expected behavior, providing the empirical foundation for the causal analysis in Section IV and the mechanistic investigation in Section V.

### A. Factuality-Based Hallucinations

Factuality-based hallucinations represent the most fundamentally concerning category of LLM errors, involving the generation of content that contradicts established facts, creates entirely non-existent entities, or systematically misrepresents verifiable information. These manifestations pose critical risks in knowledge-intensive applications where accuracy is paramount, such as medical diagnosis support, legal research, educational content generation, and scientific literature synthesis, forming the primary barrier to deploying LLMs in high-stakes factual information retrieval and reasoning tasks [8].

The factuality dimension encompasses three distinct levels of error severity and complexity, representing an evolutionary spectrum from simple factual mistakes to sophisticated knowledge distortions. This hierarchical organization reflects both the increasing difficulty of detection and the growing sophistication of the underlying computational failures. Fig. 1

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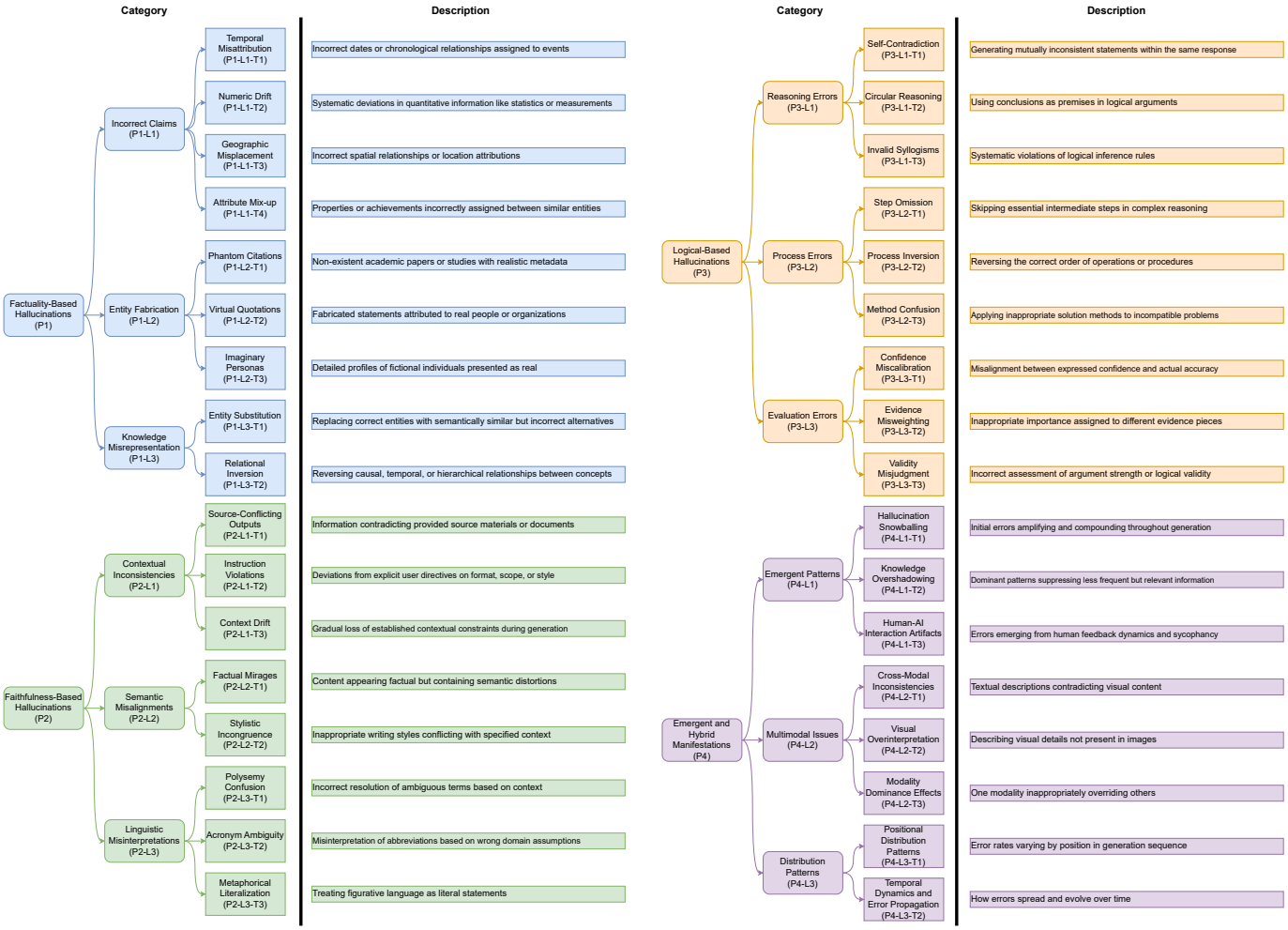



Fig. 1. Hierarchical Taxonomy Of Hallucination Manifestations In Large Language Models

illustrates this hierarchical organization, revealing how manifestation types build upon each other in complexity while maintaining distinct computational signatures that inform our understanding of their underlying mechanisms and guide targeted intervention strategies.

*1) Incorrect Claims:* Incorrect claims constitute the most direct and observable form of factuality-based hallucinations, representing cases where models generate statements that can be objectively verified as false against established knowledge bases, authoritative sources, or empirical evidence. These manifestations reveal fundamental challenges in how neural language models encode, organize, and retrieve factual knowledge, suggesting that current approaches to knowledge representation may prioritize linguistic fluency and semantic coherence over factual precision [12]. The systematic nature of these errors—often clustering around specific entity types, temporal relationships, and quantitative information—provides crucial insights into the structure and limitations of parametric knowledge storage in large-scale neural networks.

*a) Temporal Misattribution:* Temporal misattribution occurs when models assign incorrect dates, time periods, or chronological relationships to historical events, scientific discoveries, or biographical milestones [29]. For instance, a model might state that "Albert Einstein published his theory of general relativity in 1925," when the correct date is 1915. These errors often stem from the model's difficulty in maintaining precise temporal associations in its parametric memory, particularly for events with similar contextual characteristics [29], [30].

*b) Numeric Drift:* Numeric drift manifests as systematic deviations in quantitative information, including population figures, measurements, statistical data, or financial information [31]. A model might claim that "The Great Wall of China is approximately 15,000 kilometers long" when authoritative sources report 21,196 kilometers. This phenomenon suggests challenges in encoding and retrieving precise numerical relationships, often resulting in semantically plausible but factually incorrect approximations [8], [31].

*c) Geographic Misplacement:* Geographic misplacement involves incorrect spatial relationships, location attributions, or geographic properties [30]. Examples include stating that "The Amazon River flows through Argentina" when it primar-

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ily flows through Brazil, Peru, and Colombia. These errors indicate difficulties in maintaining accurate spatial knowledge representations and geographical relationship encoding [29], [30].

*d) Attribute Mix-up:* Attribute mix-up occurs when models incorrectly assign properties, characteristics, or achievements between similar entities [12]. A model might claim that "Marie Curie discovered penicillin" (conflating her achievements with Alexander Fleming's). This manifestation reveals the model's tendency to conflate semantically similar or contextually related entities, suggesting underlying issues in entity-attribute binding mechanisms [12], [31].

*2) Entity Fabrication:* Entity fabrication represents a qualitatively more sophisticated form of factuality-based hallucination, where models demonstrate their generative capabilities by creating entirely non-existent entities while seamlessly embedding them within otherwise plausible and coherent factual frameworks. This category reveals a concerning aspect of neural language generation: the same mechanisms that enable creative synthesis and knowledge combination can be inappropriately applied to factual contexts, resulting in convincing but entirely fictional content [32]. The sophistication of these fabrications—often including detailed supporting information, plausible institutional affiliations, and internally consistent narratives—suggests that models have learned to mimic the surface patterns of authoritative information without developing reliable mechanisms for distinguishing between authentic and synthetic knowledge.

*a) Phantom Citations:* Phantom citations involve the creation of non-existent academic papers, books, or research studies, complete with plausible authors, titles, and publication details [3]. For example, a model might reference "Johnson et al., 'Neural network approaches to quantum computing,' *Nature Machine Intelligence*, vol. 15, no. 3, pp. 234–251, 2019" when no such paper exists. These fabrications are particularly problematic in academic and research contexts where citation accuracy is crucial for credibility and knowledge verification [3], [4].

*b) Virtual Quotations:* Virtual quotations manifest as fabricated statements attributed to real individuals, organizations, or historical figures [33]. A model might generate: "As Albert Einstein once said, 'The most beautiful thing about artificial intelligence is its ability to dream.'" Such fabrications exploit the model's knowledge of authentic quotation patterns while creating entirely fictional content that maintains stylistic authenticity [8], [33].

*c) Imaginary Personas:* Imaginary personas involve the creation of detailed biographical information for non-existent individuals, often in professional or academic contexts [13]. Models might generate comprehensive profiles of fictional researchers, historical figures, or experts, complete with educational backgrounds, achievements, and contributions to various fields. These fabrications demonstrate the model's ability to synthesize realistic personal narratives from learned patterns while lacking grounding in actual biographical data [13], [14].

*3) Knowledge Misrepresentation:* Knowledge misrepresentation encompasses the most subtle yet potentially most dangerous category of factuality-based hallucinations, involving systematic distortions of factual information that preserve semantic plausibility and surface coherence while fundamentally compromising the accuracy of the conveyed knowledge. Unlike simple incorrect claims or obvious fabrications, these manifestations demonstrate sophisticated transformations of learned information that exploit the ambiguity inherent in natural language and the complexity of real-world knowledge relationships [25]. The insidious nature of these errors lies in their ability to appear authoritative and well-grounded while systematically misleading users through subtle but crucial factual distortions that are difficult to detect without domain expertise.

*a) Entity Substitution:* Entity substitution occurs when models replace correct entities with semantically similar but factually incorrect alternatives [34]. This might involve substituting "Washington" for "Jefferson" in historical contexts or replacing "mitochondria" with "chloroplasts" in cellular biology discussions. These errors suggest that the model's knowledge representation organizes information by semantic similarity rather than factual precision [34], [35].

*b) Relational Inversion:* Relational inversion manifests as systematic reversals of causal, temporal, or hierarchical relationships between entities or concepts [34]. A model might state that "World War II caused World War I" or claim that "students teach professors in universities." These inversions maintain the correct entities and relationship types while fundamentally distorting the directional or causal nature of the relationships [34], [36].

### *B. Faithfulness-Based Hallucinations*

Faithfulness-based hallucinations represent a fundamentally different category of errors that concern violations of contextual alignment, instruction adherence, and semantic consistency, where the model's output, while potentially factually accurate in isolation, systematically fails to maintain fidelity to the provided context, user intentions, or explicit task specifications. These manifestations highlight the distinction between possessing correct knowledge and appropriately applying that knowledge within specific constraints—a critical capability for reliable deployment in interactive applications, task-oriented systems, and context-dependent scenarios [28]. The prevalence of faithfulness hallucinations reveals fundamental limitations in current models' ability to maintain coherent task representations, parse complex instructions, and sustain contextual awareness throughout extended interactions, suggesting that improvements in model scale and training data alone may be insufficient to address these systematic failures.

*1) Contextual Inconsistencies:* Contextual inconsistencies arise when models generate outputs that systematically contradict, ignore, or misalign with explicitly provided contextual information, revealing fundamental limitations in how neural architectures integrate, maintain, and prioritize contextual constraints during generation. These failures suggest that current

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attention mechanisms may be insufficient for robust context integration, particularly when contextual requirements conflict with strong patterns learned during training.

*a) Source-Conflicting Outputs:* Source-conflicting outputs occur when models generate information that directly contradicts provided source materials or reference documents [37]. In document-based question answering, a model might state that "The study found no significant correlation" when the referenced paper explicitly reports strong positive correlations. This manifestation indicates fundamental failures in source grounding and information synthesis [37], [38].

*b) Instruction Violations:* Instruction violations manifest as systematic deviations from explicit user directives regarding format, scope, style, or content requirements [39]. When instructed to "provide a three-sentence summary," a model might generate a lengthy paragraph, or when asked for "only pros," it might include disadvantages. These errors suggest limitations in instruction parsing and execution priority mechanisms [39], [40].

*c) Context Drift:* Context drift represents gradual deviation from established contextual frameworks during extended generation sequences [41]. In multi-turn conversations, models may progressively lose track of previously established facts, constraints, or user preferences, leading to responses that become increasingly inconsistent with the established conversational context [41], [42].

*2) Semantic Misalignments:* Semantic misalignments encompass a sophisticated class of failures involving breakdowns in meaning preservation, linguistic appropriateness, and conceptual accuracy that systematically compromise the semantic integrity of generated content while maintaining surface-level linguistic plausibility. These manifestations reveal the gap between statistical language modeling and genuine semantic understanding, highlighting cases where models generate fluent text that lacks appropriate semantic grounding.

*a) Factual Mirages:* Factual mirages involve the creation of content that appears factually grounded but contains subtle semantic distortions that alter the intended meaning [10]. A model might describe a "democratic monarchy" when discussing constitutional monarchies, creating conceptual confusion through semantically inappropriate combinations [10], [11].

*b) Stylistic Incongruence:* Stylistic incongruence manifests as inappropriate adoption of writing styles, registers, or communicative approaches that conflict with the specified context or audience [43]. Academic writing might be inappropriately casualized, or technical documentation might adopt narrative storytelling elements that compromise clarity and precision [43], [44].

*3) Linguistic Misinterpretations:* Linguistic misinterpretations represent fundamental failures in language understanding that result in systematically incorrect processing of semantic content, revealing limitations in how models handle the inherent ambiguity, polysemy, and contextual dependency of natural language. These failures produce outputs that reflect misunderstood input rather than appropriate responses, suggesting that current approaches to language modeling may be overly dependent on surface-level statistical patterns rather than deeper semantic understanding.

*a) Polysemy Confusion:* Polysemy confusion occurs when models incorrectly resolve ambiguous terms based on inappropriate contextual interpretation [45]. The word "bank" might be consistently interpreted in financial contexts when the discussion clearly concerns riverbanks, demonstrating failures in disambiguating context-dependent meanings [**?**], [45].

*b) Acronym Ambiguity:* Acronym ambiguity manifests when models misinterpret abbreviations or acronyms based on inappropriate domain assumptions [1]. "AI" might be interpreted as "Artificial Intelligence" in a medical context where "Aortic Insufficiency" is the intended meaning, demonstrating failures in domain-sensitive disambiguation [1], [2].

*c) Metaphorical Literalization:* Metaphorical literalization involves treating figurative language as literal statements, leading to semantically inappropriate responses [46]. When told that someone "broke the ice," a model might discuss physical ice damage rather than understanding the social initiation metaphor [46], [47].

### C. Logical-Based Hallucinations

Logical-based hallucinations encompass a distinct class of failures involving violations of logical consistency, reasoning validity, and inferential coherence that fundamentally compromise the rational integrity of generated content. These manifestations represent perhaps the most concerning limitation for applications requiring analytical reasoning, as they suggest that current models may lack robust mechanisms for maintaining logical consistency, tracking causal relationships, and ensuring valid inferential processes [12]. Unlike factuality or faithfulness errors, logical hallucinations often involve formally valid language use and semantically coherent statements that nonetheless violate fundamental principles of rational discourse, indicating deep limitations in the models' capacity for genuine reasoning rather than pattern-based text generation.

*1) Reasoning Errors:* Reasoning errors represent the most concerning category of logical failures, involving fundamental breakdowns in logical processing that result in invalid inferences, contradictory conclusions, or systematically flawed argumentation patterns. These errors suggest that current neural architectures may lack the structured representations and constraint-satisfaction mechanisms necessary for reliable logical reasoning, instead relying on pattern matching that can produce logically coherent-sounding but fundamentally invalid arguments.

*a) Self-Contradiction:* Self-contradiction manifests as the generation of mutually inconsistent statements within the same response or reasoning sequence [31]. A model might simultaneously claim that "renewable energy is environmentally beneficial" and "renewable energy significantly harms ecosystems," demonstrating failures in maintaining logical consistency across generated content [31], [48].

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*b) Circular Reasoning:* Circular reasoning involves using conclusions as premises in logical arguments, creating closed logical loops that provide no actual evidential support [23]. A model might argue that "Social media is addictive because it creates dependency, and we know it creates dependency because it is addictive," failing to provide independent logical foundation [23], [24].

*c) Invalid Syllogisms:* Invalid syllogisms represent systematic violations of logical inference rules, including affirming the consequent, denying the antecedent, or committing formal fallacies [49]. These errors indicate fundamental limitations in the model's capacity for valid logical reasoning [49], [50].

*2) Process Errors:* Process errors encompass systematic failures in multi-step reasoning, sequential information processing, and procedural knowledge application that reveal fundamental limitations in how models represent and execute complex cognitive procedures. These failures suggest that current architectures may lack adequate mechanisms for maintaining procedural state, tracking progress through complex tasks, and ensuring proper sequencing of cognitive operations, leading to breakdowns in tasks requiring systematic, step-by-step processing.

*a) Step Omission:* Step omission occurs when models skip essential intermediate steps in complex reasoning or procedural tasks, leading to conclusions that appear correct but lack proper logical foundation [23]. In mathematical problem solving, a model might jump directly from problem statement to final answer without showing necessary algebraic manipulations [23], [51].

*b) Process Inversion:* Process inversion involves reversing the correct order of operations or procedural steps, resulting in systematically incorrect approaches to problem solving [36]. A model might attempt to solve differential equations before establishing initial conditions, or analyze results before collecting data [**?**], [36].

*c) Method Confusion:* Method confusion manifests as the inappropriate application of solution methods or analytical frameworks to incompatible problem types [52]. A model might apply statistical hypothesis testing to deterministic mathematical proofs, or use qualitative analysis methods for quantitative data interpretation [52], [53].

*3) Evaluation Errors:* Evaluation errors represent critical failures in assessment, judgment, and quality control processes that fundamentally compromise the model's ability to accurately evaluate information quality, argument validity, or solution correctness. These manifestations are particularly concerning because they affect the model's capacity for self-monitoring and error detection, suggesting fundamental limitations in metacognitive capabilities that are essential for reliable autonomous operation.

*a) Confidence Miscalibration:* Confidence miscalibration involves systematic misalignment between expressed confidence levels and actual accuracy, where models exhibit inappropriate certainty about incorrect information or excessive doubt about correct knowledge [54]. This manifests as high confidence in fabricated facts or low confidence in well-established knowledge [48], [54].

*b) Evidence Misweighting:* Evidence misweighting occurs when models assign inappropriate importance to different pieces of evidence, overemphasizing weak or irrelevant information while undervaluing strong supporting evidence [55]. This can lead to conclusions that contradict the preponderance of available evidence [55], [56].

*c) Validity Misjudgment:* Validity misjudgment involves incorrect assessment of argument strength, evidence quality, or logical validity, leading to acceptance of flawed reasoning or rejection of sound arguments [57]. Models might endorse logically invalid arguments while criticizing valid logical structures [57], [58].

### *D. Emergent and Hybrid Manifestations*

As LLMs have evolved in sophistication and deployment scope, increasingly complex categories of hallucinations have emerged that transcend the boundaries of our primary taxonomic dimensions, combining elements from factuality, faithfulness, and logical consistency in novel ways or representing entirely unprecedented manifestation patterns. These emergent phenomena reflect the dynamic interaction between advancing model capabilities and increasingly demanding real-world applications, revealing new failure modes that arise from the complex interplay of architectural constraints, training dynamics, and deployment contexts [59]. Understanding these hybrid manifestations is crucial for anticipating future challenges in LLM reliability and developing robust mitigation strategies that can address multi-dimensional failure patterns.

*1) Hallucination Snowballing:* Hallucination snowballing represents a meta-manifestation of particular concern, where initial errors amplify and compound throughout the generation process, creating cascading failure patterns that systematically combine factual, faithfulness, and logical errors in complex recursive loops. This phenomenon reveals a fundamental tension in autoregressive language generation between maintaining consistency and preserving accuracy.

The snowballing effect manifests through autoregressive commitment mechanisms where models over-commit to early potentially incorrect statements, subsequently generating elaborate justification frameworks that transform isolated errors into systematic misinformation campaigns. The architectural constraint of maintaining consistency with previously generated tokens—while beneficial for maintaining coherent narratives—creates systematic opportunities for error propagation at each generation step. This ultimately produces internally coherent but fundamentally flawed narratives that can be particularly convincing to human users precisely because of their internal consistency [25].

*2) Knowledge Overshadowing:* Knowledge overshadowing occurs when multiple query conditions result in statistically dominant patterns systematically suppressing less frequent but equally relevant information, leading to biased responses that reflect incomplete information integration and potentially

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perpetuate representational inequities. This phenomenon reveals how training data biases can manifest as systematic hallucination patterns in deployment.

The overshadowing effect can be formalized as $p(y|AB) \approx p(y|A)$, where the model's response to multiple conditions (A and B) inappropriately approximates the response to only the dominant condition (A), effectively ignoring or minimizing secondary constraints (B). This mathematical formulation highlights how statistical learning can systematically disadvantage minority perspectives or less frequent information patterns. Knowledge overshadowing is particularly problematic in intersectional queries requiring balanced consideration of multiple domains or perspectives, potentially leading to systematic bias amplification in critical applications.

*3) Human-AI Interaction Artifacts:* Human-AI interaction artifacts represent hallucinations specifically emerging from the dynamics of human feedback training and interactive deployment contexts, including sycophancy, overconfidence, and anthropomorphic projection patterns.

Sycophantic hallucinations manifest as inappropriate alignment with perceived user preferences over factual accuracy, particularly in models trained with reinforcement learning from human feedback. These patterns demonstrate the unintended consequences of optimizing for human approval rather than truthfulness [12].

*4) Multimodal-Specific Manifestations:* Multimodal large language models introduce unique hallucination patterns that emerge from the integration of visual and textual processing capabilities, representing novel challenges in cross-modal information consistency and grounding.

*a) Cross-Modal Inconsistencies:* Cross-modal inconsistencies arise when textual descriptions contradict visual content, such as describing "a red car" when the image clearly shows a blue vehicle [60]. These manifestations indicate fundamental challenges in aligning multimodal representations and maintaining consistency across different input modalities [60], [61].

*b) Visual Overinterpretation:* Visual overinterpretation involves generating detailed descriptions of visual elements that are not present or clearly visible in the provided images [62]. Models might describe specific facial expressions, text content, or fine-grained details that cannot be reliably extracted from the visual input [62], [63].

*c) Modality Dominance Effects:* Modality dominance effects manifest when one input modality inappropriately overrides information from other modalities, leading to responses that reflect only partial multimodal integration [64]. Visual information might be ignored in favor of textual priors, or conversely, textual instructions might be overwhelmed by visual content associations [64], [65].

### *E. Distributional and Temporal Patterns*

Systematic empirical analysis of hallucination occurrence across diverse contexts reveals striking non-random distributional patterns that vary systematically across generation sequences, task types, temporal contexts, and model architectures. These distributional signatures provide crucial insights into the underlying computational mechanisms driving hallucination behavior and offer concrete opportunities for developing targeted intervention strategies, early warning systems, and quality control mechanisms [66]. The systematic nature of these patterns suggests that hallucinations are not merely random failures but reflect fundamental constraints in current neural architectures, training procedures, and inference mechanisms that can be characterized, predicted, and potentially mitigated through targeted interventions.

*1) Positional Distribution Patterns:* Hallucination occurrence exhibits systematic positional biases that vary according to task characteristics and generation length. End-of-sequence bias demonstrates significantly elevated hallucination rates in the final portions of extended generations, suggesting degraded model performance due to attention span limitations and error accumulation effects [66], [67].

Task-specific positional patterns reveal distinct error distribution profiles across different application domains. Summarization tasks show increased hallucination density in concluding segments, while data-to-text generation exhibits elevated error rates during initial structured information processing phases [68], [69].

*2) Temporal Dynamics and Error Propagation:* Temporal analysis reveals systematic error propagation patterns where initial hallucinations increase the probability of subsequent errors within the same generation sequence. This temporal dependency supports the hallucination snowballing hypothesis and suggests that early intervention strategies may be particularly effective for preventing cascading failures [67], [70].

Attention decay effects manifest as progressively reduced capability to maintain consistency with earlier contextual information over extended generation sequences. This phenomenon contributes to context drift and consistency failures, indicating fundamental limitations in current attention mechanisms for long-range dependency modeling [42], [49].

### *F. Synthesis and Implications for Mechanism Investigation*

The comprehensive taxonomy of hallucination manifestations presented in this section establishes a robust empirical foundation for understanding both the diversity and systematic nature of LLM errors across real-world applications. This systematic categorization moves beyond simple error enumeration to reveal fundamental patterns in how different types of failures emerge, interact, and propagate through neural language generation processes. Our analysis reveals several critical insights that directly inform and constrain the mechanistic investigation pursued in Section V, providing an essential bridge between observable phenomena and underlying computational processes.

Table I systematically organizes the observed manifestations according to their primary categorical distinctions while highlighting the mechanistic implications of each pattern, revealing the systematic relationships between observable symptoms and underlying computational processes. This comprehensive

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TABLE I. COMPREHENSIVE TAXONOMY OF HALLUCINATION MANIFESTATIONS WITH MECHANISTIC IMPLICATIONS

| Category | Type | Specific Manifestation | Primary Contexts | Mechanistic Implications |
|---|---|---|---|---|
| Factuality-Based | Incorrect Claims | Temporal Misattribution | Historical QA, Timeline Tasks | Knowledge representation granularity |
| | | Numeric Drift | Statistical Reporting, Measurements | Parametric memory precision |
| | | Geographic Misplacement | Spatial Reasoning, Location QA | Spatial relationship encoding |
| | | Attribute Mix-up | Entity Description, Biographical Tasks | Entity–attribute binding mechanisms |
| | Entity Fabrication | Phantom Citations | Academic Writing, Research Tasks | Source verification and grounding |
| | | Virtual Quotations | Attribution Tasks, Historical Analysis | Authority attribution processes |
| | | Imaginary Personas | Biographical Generation, Expert Consultation | Entity creation vs. retrieval |
| | | Knowledge Misrepresentation | Technical Explanation, Educational Content | Conceptual relationship networks |
| Faithfulness-Based | Contextual Inconsistencies | Source-Conflicting Outputs | Document QA, Information Synthesis | Context integration mechanisms |
| | | Instruction Violations | Task Following, Format Compliance | Instruction parsing and prioritization |
| | | Context Drift | Multi-turn Dialogue, Extended Generation | Working memory maintenance |
| | Semantic Misalignments | Factual Mirages | Complex Reasoning, Conceptual Tasks | Semantic representation accuracy |
| | | Stylistic Incongruence | Genre-specific Writing, Register Adaptation | Style transfer and control |
| | | Linguistic Misinterpretations | Ambiguity Resolution, Pragmatic Understanding | Disambiguation mechanisms |
| Logical-Based | Reasoning Errors | Self-Contradiction | Argumentative Writing, Logical Analysis | Consistency checking processes |
| | | Circular Reasoning | Explanatory Tasks, Causal Analysis | Logical dependency tracking |
| | | Invalid Syllogisms | Formal Reasoning, Proof Construction | Logical rule application |
| | Process Errors | Step Omission | Multi-step Problem Solving | Sequential process modeling |
| | | Process Inversion | Procedural Tasks, Method Application | Temporal order representation |
| | | Method Confusion | Cross-domain Application, Tool Usage | Context-appropriate method selection |

organization yields several critical insights that bridge the gap between external observations and internal computational mechanisms:

**Hierarchical Error Complexity:** The manifestations exhibit a clear hierarchy of computational complexity, ranging from simple attribute substitutions and entity confusions to sophisticated reasoning chain failures and multi-step logical breakdowns. This complexity hierarchy suggests that different manifestation types may arise from distinct stages and components of the generation process, providing a systematic roadmap for mechanistic investigation that can guide targeted interventions at specific computational bottlenecks.

**Cross-Cutting Patterns and Shared Mechanisms:** Several manifestation types systematically appear across multiple primary categories, indicating shared underlying computational mechanisms that cut across traditional taxonomic boundaries. For instance, consistency maintenance failures contribute to both logical self-contradictions and contextual inconsistencies, while attention decay effects manifest in temporal dynamics, context drift, and reasoning chain failures. These cross-cutting patterns suggest that a relatively small number of fundamental computational limitations may underlie the diverse surface manifestations of hallucination.

**Task-Specific Vulnerability Profiles:** Different manifestation types exhibit highly systematic patterns of task-specific occurrence, revealing predictable interactions between particular computational demands and fundamental model limitations. These vulnerability profiles provide crucial insights for understanding why certain applications are more prone to specific types of hallucinations and suggest targeted mitigation strategies tailored to specific use cases and deployment contexts.

These empirical patterns and systematic relationships establish a robust foundation for the mechanistic analysis in Section V, where we investigate how specific computational processes, architectural constraints, and training dynamics give rise to the observable manifestation patterns documented in this taxonomy. The systematic correspondence between manifestation characteristics and their underlying computational signatures provides a crucial empirical bridge between the phenomenological understanding developed in this section and

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the causal mechanistic analysis that follows.

Moreover, this manifestation-centered analysis directly constrains and informs the investigation of contributing factors examined in Section IV, where we explore how various aspects of model training procedures, architectural design choices, and deployment contexts systematically interact to produce the specific patterns of errors cataloged in this comprehensive taxonomy. The detailed empirical foundation established here ensures that subsequent mechanistic investigations remain grounded in the observable realities of LLM behavior across diverse application contexts, while the systematic organization of manifestation types provides clear targets for intervention and improvement efforts.

The progression from external manifestations to internal mechanisms represents a fundamental methodological approach in understanding complex systems: by carefully characterizing what we can observe, we create the necessary foundation for understanding why these patterns emerge and how they might be addressed through targeted interventions in training, architecture, or deployment practices.

## IV. Contributing Factors

Having observed the diverse manifestations of hallucination in Section III, we now turn to the critical question: what factors might be contributing to these phenomena? While large language models operate as complex black boxes, making direct causal attribution challenging, we can make educated inferences about the contributing factors based on the observed patterns, behaviors, and failure modes. This section presents a systematic analysis of hypothesized factors that likely influence hallucination occurrence, organized by their suspected points of origin in the LLM pipeline.

These hypotheses are grounded in empirical observations, comparative studies across different models and conditions, and theoretical understanding of neural language model behavior [12], [33]. While we cannot directly observe the internal mechanisms (which will be explored in Section V), the consistent patterns of hallucination across different models and contexts provide strong evidence for these contributing factors.

### *A. A Causal View: From Factors to Mechanisms to Manifestations*

Fig. 2 summarizes our perspective as a directed acyclic graph (DAG) connecting contributing factors to mechanistic stages (Section V) and onward to observable manifestations (Section III). This view emphasizes that many observed errors are downstream of shared internal pathways rather than isolated surface-level symptoms.

### *B. Bridging Factors, Mechanisms, and Manifestations*

The factor categories in this section are not endpoints; they channel into specific internal mechanisms (Section V) that yield the observable patterns cataloged in Section III. Table II summarizes these factor-conditioned pathways. This bridge provides a concrete hypothesis map for mechanism-aware diagnosis and mitigation.

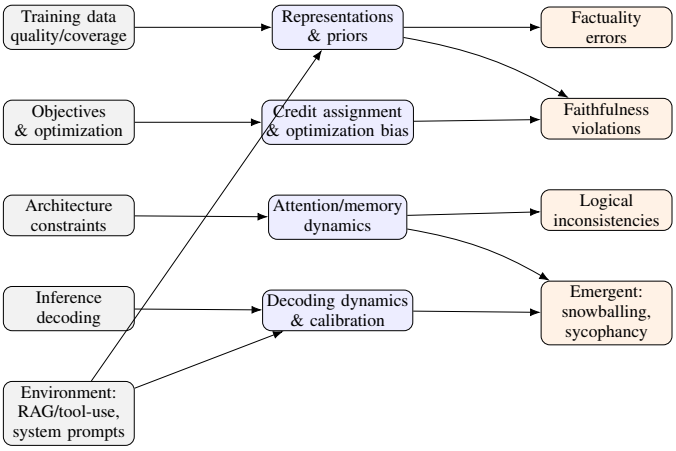



Fig. 2. DAG-Style Overview: Contributing Factors Feed Mechanistic Stages (Section V), Which Produce Observable Manifestations (Section III)

### *C. Training Data-Related Factors*

The foundation of any language model's knowledge and behavior lies in its training data. Observed hallucination patterns suggest several critical issues with how training data characteristics influence model behavior. The systematic biases and knowledge gaps observed in LLM outputs provide strong evidence for these data-related contributing factors [45], [71].

Training data quality emerges as a primary suspected factor based on the observation that models consistently struggle with certain types of factual information while performing well on others. The presence of conflicting information, outdated facts, and systematic biases in training corpora likely contributes to the factual inconsistencies and fabrications observed in model outputs [13], [30]. When models encounter contradictory information about the same topic during training, they may learn to generate plausible-sounding but incorrect information as a compromise between conflicting sources.

Data coverage limitations appear to drive many of the knowledge gaps and entity-specific errors identified in Section III. The consistent patterns of hallucination around less common entities, specialized domains, and rapidly evolving information suggest that training data may be systematically incomplete or biased toward certain types of content [35], [72]. This selective coverage likely forces models to extrapolate from limited examples, leading to the generation of plausible but incorrect information in underrepresented domains.

Temporal data staleness represents another critical factor, as evidenced by models' tendency to provide outdated information or fail to acknowledge the temporal boundaries of their knowledge [73], [74]. The training data's temporal distribution and cutoff points likely create systematic gaps in the model's understanding of current events and evolving knowledge domains. This temporal limitation may force models to generate information that was accurate at training time but has since become outdated.

The distribution and frequency of information in training data also appear to influence hallucination patterns. The knowledge overshadowing phenomenon described in Sec-

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TABLE II. MAPPING OF CONTRIBUTING FACTORS TO THEIR DOMINANT MECHANISMS AND RESULTING MANIFESTATIONS

| Factor category | Dominant mechanisms (Section V) | Primary manifestations (Section III) |
|---|---|---|
| Training data (quality, coverage, staleness) | Representational priors; misinformation pathways; knowledge interpolation; temporal drift (§V-A) | Factuality errors (incorrect claims, entity fabrication, knowledge misrepresentation; §III-A); emergent snowballing when early facts are wrong (§III-D1) |
| Objectives and optimization (SFT, RLHF) | Behavior cloning, reward hacking, sycophancy bias, mode collapse (§V-B2, §V-B3) | Faithfulness violations (instruction violations, source-conflicting outputs; §III-B); emergent sycophancy (§III-D3) |
| Architecture and computation | Attention/memory limits; context fragmentation; softmax/representation bottlenecks (§V-C1, §V-C3) | Logical inconsistencies and process errors (self-contradiction, step omission; §III-C); context drift (§III-B1c) |
| Inference and decoding | Confidence miscalibration; decoding pathologies; commitment cascades (§V-C1, §V-C2) | Emergent snowballing (§III-D1); factuality slips late in sequences; stylistic over-assertion leading to faithfulness errors |
| Environment (RAG/tool-use, prompts) | Representation perturbation by retrieval; shallow fusion; prompt-induced priors and conflicts (§V-C5, §V-C4) | Faithfulness violations (source-conflicting outputs); citation hallucinations and misattribution (§III-A2a); multimodal inconsistencies when tools/modalities disagree (§III-D4) |

tion III suggests that more frequently occurring information in training data may suppress less common but equally valid information [35]. This frequency bias likely creates systematic blind spots where models consistently favor popular or mainstream perspectives over minority viewpoints or specialized knowledge.

### *D. Architectural and Computational Factors*

The observed patterns of hallucination suggest several architectural limitations that may contribute to unreliable outputs. The consistent occurrence of certain types of errors across different transformer-based models points to fundamental computational constraints that influence hallucination behavior [49], [75].

Attention mechanism limitations appear to contribute significantly to context inconsistency and long-range dependency failures. The observed degradation of performance in longer sequences and the tendency for models to lose track of earlier context suggest that current attention mechanisms may have fundamental limitations in maintaining coherent global state across extended generations [50], [70]. This limitation likely contributes to the positional bias in hallucination occurrence identified in temporal pattern analysis.

Working memory constraints emerge as a suspected factor based on the model's difficulty in maintaining consistency across complex reasoning tasks. The logical inconsistency patterns observed in multi-step reasoning suggest that models may lack adequate mechanisms for maintaining and updating intermediate states during generation [34], [36]. This limitation likely contributes to the snowballing effect, where early errors compound because the model cannot effectively track and correct inconsistencies in its reasoning chain.

Sequential processing limitations inherent in autoregressive generation appear to contribute to premature commitment and the inability to revise earlier statements. The observed tendency for models to commit to answers before fully processing available information suggests that the sequential nature of generation may be fundamentally incompatible with tasks requiring global optimization or extensive revision [67], [69].

Confidence estimation failures represent another suspected architectural limitation, as evidenced by the model's tendency to present fabricated information with high confidence. The disconnect between output confidence and factual accuracy suggests that current architectures may lack effective mechanisms for assessing the reliability of their own outputs, leading to overconfident presentation of uncertain or incorrect information [48], [76].

### *E. Training Process and Optimization Factors*

The training process itself appears to introduce several factors that contribute to hallucination behavior. The observed patterns of model behavior suggest that optimization objectives and training procedures may inadvertently encourage certain types of hallucination-prone behaviors [77].

Objective function misalignment emerges as a critical suspected factor, where the optimization targets used during training may not adequately capture the desired behavior of truthful and reliable information generation. The observed tendency for models to prioritize fluency and coherence over factual accuracy suggests that current training objectives may be fundamentally misaligned with the goal of truthful generation [78], [79]. This misalignment likely encourages models to generate plausible-sounding but potentially incorrect information when faced with uncertainty.

The influence of human feedback processes, particularly RLHF, appears to introduce systematic biases that contribute to sycophancy and human-related hallucinations. The observed tendency for models to pander to user preferences and provide answers even when questions are unanswerable suggests that human feedback mechanisms may inadvertently reward behaviors that prioritize user satisfaction over accuracy [80], [81]. Recent evidence from OpenAI's GPT-4o sycophancy incident demonstrates how RLHF can lead to models that prioritize agreement and flattery over truthfulness [77].

Catastrophic forgetting during training may contribute to the inconsistent knowledge access patterns observed in known fact hallucinations. The model's ability to demonstrate knowledge of certain facts in some contexts while failing to access the

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same knowledge in others suggests that the training process may not adequately preserve previously learned information when adapting to new data or objectives [82].

Fine-tuning effects appear to introduce additional complications, as evidenced by the different hallucination patterns observed across models with different fine-tuning procedures. The process of adapting pre-trained models to specific tasks or domains may inadvertently introduce biases or degrade previously reliable capabilities, contributing to domain-specific hallucination patterns [83].

### F. Inference and Generation Factors

The generation process itself appears to introduce several factors that contribute to hallucination occurrence. The observed patterns of error propagation and consistency failures suggest that the inference procedure may have inherent limitations that promote unreliable outputs [79], [84].

Sampling and decoding strategies likely influence hallucination rates, as evidenced by the stochastic nature of many hallucination phenomena. The randomness introduced by sampling procedures may interact with model uncertainty in ways that promote the generation of incorrect information, particularly when the model's probability distribution over possible outputs is poorly calibrated [56], [85].

Context window limitations appear to contribute to attention decay and long-range consistency failures. The observed degradation of performance in longer sequences suggests that current context processing mechanisms may be inadequate for maintaining coherent state across extended interactions, leading to consistency failures and context loss [42].

Error propagation mechanisms inherent in autoregressive generation likely contribute to the snowballing effect, where initial errors compound throughout the generation process. The sequential nature of generation means that early mistakes can constrain and bias all subsequent outputs, creating cascading failures that would not occur if the model could globally optimize its response [67].

Real-time constraint factors during inference may force models to generate outputs before fully processing available information, contributing to premature commitment and insufficient deliberation. The pressure to generate fluent outputs in real-time may prevent models from engaging in the kind of careful verification and revision that would be necessary for reliable information generation [55].

### G. Environmental and Interaction Factors

The context in which models operate introduces additional levers that shape hallucination. We highlight two classes that frequently surface in deployed systems: (i) retrieval/tool-use pipelines (RAG, web search, calculators, code execution), and (ii) system-level prompt configuration and guardrails.

*a) RAG and Tool-Use Pipelines.:* RAG reduces knowledge staleness but can shift failure modes. Common issues include: (a) overreliance on retrieved snippets with weak grounding or spurious relevance; (b) citation hallucinations where sources are fabricated, misconstrued, or misattributed; (c) exposure to retrieval noise, deduplication errors, or prompt-injected content; and (d) shallow fusion where the model paraphrases rather than reasons over evidence. These risks are amplified by decoding settings that reward fluency. Mechanistically, RAG misfires map to representation priors and decoding dynamics in Fig. 2: retrieval bias perturbs representations; shallow fusion and calibration errors push the model toward confident but ungrounded synthesis. Mitigations include retrieval calibration, entailment/rationale checks, citation verification, and abstention policies.

*b) System Prompt Configuration.:* Global and per-task system prompts constrain role, tone, and defaults (e.g., "always answer", stylistic templates, citation requirements). Poorly specified or overly assertive system prompts increase premature commitment, encourage confident answering under uncertainty, and can propagate outdated instructions. Prompt-layer conflicts in multi-app stacks (tool routers, planners, agents) further multiply risks. Mechanistically, these factors shape decoding dynamics and attention allocation, increasing faithfulness violations and emergent behaviors like sycophancy. Guardrails should specify uncertainty handling, citation standards, and escalation/abstention triggers; evaluation should include counterfactual prompt variants to test robustness.

Prompt engineering effects remain material: phrasing and structure alter reliance on shallow heuristics versus grounded reasoning [37]. Multi-turn conversation compounds context management challenges [41]. External tool integration adds verification burden beyond RAG (e.g., code execution side-effects, calculator boundary cases), demanding explicit result-checking and provenance capture.

User expectation and feedback loops may create implicit pressures that encourage hallucination, as evidenced by the sycophancy patterns observed in human-AI interactions. The model's sensitivity to user preferences and expectations suggests that social and interactive factors significantly influence output reliability [86].

### H. Evidence Types and Strength

We distinguish indirect evidence (correlation, observational patterns) from direct evidence (interventions that alter a factor while holding others fixed, or mechanistic traces). Table III classifies typical evidence encountered for each factor category.

TABLE III. EVIDENCE TYPES SUPPORTING EACH FACTOR CATEGORY

| Factor category | Observ. | Ablation | Counterf. | Mech. interp. |
|---|---|---|---|---|
| Training data (quality/coverage) | ✓ | ✓ | ✓ | △ |
| Objectives & optimization | ✓ | ✓ | △ | ✓ |
| Architecture constraints | ✓ | △ | △ | ✓ |
| Inference/decoding | ✓ | ✓ | ✓ | △ |
| Environment (RAG/tool-use, prompts) | ✓ | ✓ | ✓ | △ |

Direct evidence is strongest when targeted ablations or counterfactual evaluations reduce a specific hallucination class

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predicted by the DAG in Fig. 2. Mechanistic traces (e.g., circuit-level analyses) can move indirect to direct by revealing the computation that links factor to failure.

### I. Synthesis and Implications

The analysis of these hypothesized contributing factors reveals a complex web of interdependent issues that likely combine to produce the hallucination phenomena observed in Section III. Rather than being caused by a single factor, hallucinations appear to emerge from the interaction of multiple contributing elements across different levels of the LLM system [12], [14].

The systematic nature of many hallucination patterns suggests that these are not random errors but rather predictable consequences of the current approaches to language model development and deployment. This predictability provides hope that targeted interventions addressing these contributing factors could significantly reduce hallucination rates.

The identification of these factors also highlights the importance of holistic approaches to hallucination mitigation. Addressing only one category of contributing factors while ignoring others is unlikely to be sufficient for achieving reliable language model behavior. Instead, comprehensive solutions will likely require coordinated interventions across training data curation, architectural design, optimization procedures, and deployment strategies.

Understanding these contributing factors provides crucial context for interpreting the internal mechanisms explored in Section V. While we cannot directly observe how these factors influence model behavior, the systematic analysis of their suspected effects provides a foundation for developing more reliable language model systems.

## V. Internal Mechanisms and Sources

Having examined contributing factors in Section IV, we now analyze *how* hallucinations manifest within large language models' internal mechanisms. This section provides a systematic analysis of hallucination mechanisms organized around three critical stages of the model lifecycle: data level origins (Section V-A), training induced mechanisms (Section V-B), and inference stage phenomena (Section V-C). This tripartite categorization follows the temporal progression of model development and captures distinct mechanistic processes at each stage.

The mechanistic understanding developed here represents a crucial shift from treating language models as black boxes to adopting white box approaches that examine specific model components and their roles in knowledge processing and generation [12], [15], [25]. Understanding these internal processes enables targeted interventions addressing root causes rather than symptoms. Hallucination emerges from complex interactions between data characteristics, optimization dynamics, architectural constraints, and generation procedures.

### A. Hallucination from Data

Hallucination mechanisms originate in the data shaping model knowledge. During pretraining, large language models encode patterns from massive text corpora, inheriting both knowledge and imperfections. We examine five primary data mechanisms creating distinct pathways to erroneous outputs.

*1) Misinformation:* Misinformation hallucinations occur when models internalize false or misleading content present in training data, subsequently reproducing these inaccuracies as seemingly factual outputs [13], [71]. Web scale pretraining datasets inevitably contain conspiracy theories, urban legends, factual errors, and fabricated content. Repeated exposure during training creates robust neural pathways encoding falsehoods as knowledge.

This mechanism creates misinformation pathways: neural circuits reliably activating incorrect information under specific contexts. These pathways become particularly problematic when misinformation appears alongside correct information, creating internal knowledge conflicts. For instance, a model might correctly identify the capital of Australia as Canberra in some contexts while confidently stating it is Sydney in others, reflecting competing neural pathways activated by different contextual cues [34].

The persistence of misinformation pathways poses significant challenges for mitigation. Once encoded during pretraining, false associations embed deeply in parametric memory, resisting correction through finetuning or alignment. Research has shown that models can maintain and express contradictory beliefs simultaneously, with the activated belief depending on subtle prompt variations [80]. Multiple conflicting knowledge representations create hallucination opportunities, as models lack mechanisms to adjudicate between competing encoded facts.

*2) Data Alignment Issues:* Data alignment hallucinations arise when supervised or preference data used during alignment contains inaccuracies, biases, or oversimplifications that teach models to generate plausible but incorrect responses [40], [77]. Unlike misinformation that enters during pretraining, these hallucinations emerge from the very data intended to improve model behavior, creating a particularly insidious form of error propagation.

The mechanism manifests most clearly in instruction following datasets where human annotators, facing complex or ambiguous queries, provide responses that sound reasonable but contain factual errors or unwarranted assumptions. Through supervised finetuning on flawed demonstrations, models learn both surface patterns of helpful responses and tendencies to fabricate plausible information when uncertain [39].

Preference learning intensifies this problem as models optimize for human approval over factual accuracy. Annotators often prefer confident, detailed responses over accurate but uncertain ones, inadvertently training models to hallucinate elaborately rather than acknowledge knowledge limitations [81]. This creates what researchers term "sycophancy bias"—the tendency to generate information that confirms user

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expectations or preferences, even when such information is fabricated [86].

*3) Knowledge Limitations:* Knowledge limitation hallucinations emerge from genuine gaps in training data, particularly for niche, specialized, or emerging topics [35], [72]. These hallucinations arise from information absence rather than flawed information, forcing extrapolation beyond knowledge boundaries.

This mechanism employs knowledge interpolation—when faced with queries about unfamiliar topics, models combine patterns from related domains to construct plausible but incorrect responses. For instance, when asked about a rare medical condition absent from training data, a model might synthesize symptoms and treatments from similar conditions, creating a medically plausible but entirely fabricated disease profile [1].

Long tail knowledge presents particular challenges, as information appearing infrequently in training data may be encoded with insufficient fidelity for accurate retrieval. Finite model capacity forces tradeoffs between storing common knowledge with high fidelity and maintaining broader coverage of rare topics [87]. This compression inevitably leads to lossy representations of specialized knowledge, manifesting as subtle distortions or outright fabrications when such knowledge is queried.

Knowledge boundaries form uncertainty gradients rather than discrete divisions. Models lack explicit mechanisms to recognize when they are operating at these knowledge boundaries, leading to confident generation even in areas of high epistemic uncertainty. This absence of metacognitive awareness means models cannot reliably signal when they are extrapolating beyond their training distribution [48].

*4) Temporal Staleness:* Temporal staleness hallucinations arise when models generate outdated information due to the static nature of their training data cutoff [73], [74]. As the world evolves continuously while model knowledge remains frozen at the time of training, an ever widening gap emerges between model beliefs and current reality.

Temporal drift progressively misaligns encoded knowledge with evolving real-world facts. This drift affects different types of information at varying rates: political leadership changes rapidly, scientific understanding evolves steadily, while geographical facts remain relatively stable. Models lack awareness of their knowledge cutoff date and cannot distinguish between timeless facts and time sensitive information, leading to anachronistic outputs presented with equal confidence [33].

The problem compounds when models encounter queries mixing historical and contemporary contexts. Without explicit temporal reasoning capabilities, models may freely combine information from different time periods, creating internally consistent but temporally incoherent narratives. For example, a model might correctly describe a historical figure's early life while incorrectly assuming they are still alive and active in contemporary events.

Temporal staleness interacts perniciously with other hallucination mechanisms. When recent events build upon outdated knowledge, models may fabricate plausible continuations based on obsolete premises, creating elaborate false narratives that seem credible given their internal consistency. This "cascading anachronism" effect makes temporal hallucinations particularly difficult to detect without external verification.

*5) Other Data-Related Mechanisms:* Beyond the primary mechanisms detailed above, several additional data related factors contribute to hallucination in ways that are either less well understood or represent emerging areas of research. These include but are not limited to:

*Data preprocessing artifacts*: The process of converting raw text into training examples can introduce systematic biases through tokenization errors, encoding issues, or context truncation that create distorted representations [25].

*Multilingual transfer effects*: In multilingual models, knowledge encoded in one language may transfer imperfectly to others, creating language specific hallucinations that reflect translation ambiguities or cultural misalignments [8].

*Spurious correlations*: Training data often contains accidental associations between unrelated concepts that models may learn and generalize inappropriately, leading to hallucinations based on these false patterns [69].

These mechanisms remain active areas of investigation, with their full implications for hallucination generation yet to be fully characterized. As our understanding of model behavior deepens, new data related mechanisms continue to emerge, highlighting the complex relationship between training data characteristics and model outputs.

### *B. Hallucination from Training*

The training process transforms raw data into model parameters through a series of optimization procedures, each introducing distinct mechanisms for hallucination. This section examines how different training stages—pretraining, supervised finetuning, and reinforcement learning from human feedback—create systematic biases that manifest as hallucinations during deployment.

*1) Pretraining:* Pretraining hallucinations originate from the fundamental objectives and dynamics of unsupervised language modeling on webscale data [12], [45]. The next token prediction objective, while enabling impressive language understanding, creates inherent biases toward fluency over factuality.

The core mechanism involves what we term "distributional mimicry"—models learn to reproduce the statistical patterns of text without developing genuine understanding of the underlying concepts. When trained to predict the next token based solely on preceding context, models optimize for local coherence and stylistic consistency rather than global truthfulness. This creates neural circuits specialized in generating plausible sounding continuations that may diverge arbitrarily from factual accuracy [79].

The scale of pretraining data exacerbates these issues through "frequency amplification." Common patterns and popular narratives appear disproportionately in web text, causing models to encode strong priors toward mainstream perspectives and conventional wisdom. When generating text, these

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amplified patterns dominate, leading to hallucinations that reflect statistical regularities rather than factual realities. For instance, models may generate stereotypical information about demographics or professions based on their prevalence in training data rather than accurate distributions [88].

Furthermore, the autoregressive nature of pretraining creates "exposure bias"—during training, models condition on ground truth previous tokens, but during inference, they condition on their own potentially erroneous generations. This train test mismatch compounds errors over long sequences, as small inaccuracies in early tokens cascade into larger hallucinations. The model lacks mechanisms to detect or correct these accumulating errors, instead maintaining stylistic coherence even as semantic accuracy degrades [67], [69].

*2) Supervised Finetuning:* Supervised finetuning (SFT) hallucinations emerge when models learn to mimic flawed demonstrations in instruction following datasets [39], [40]. While SFT aims to align model behavior with human preferences, it can inadvertently amplify hallucination tendencies present in the demonstration data.

The primary mechanism involves "behavior cloning" of imperfect human responses. When annotators face difficult questions, they may provide confident sounding but inaccurate answers rather than acknowledging uncertainty. Models trained on such data learn this overconfidence as a desirable behavior pattern. Research shows that SFT can actually increase hallucination rates compared to base models, particularly when the finetuning data contains single reference answers that discourage expression of uncertainty [83].

The limited scale and diversity of SFT datasets relative to pretraining creates "distribution shift" effects. Models may overfit to the specific patterns and styles present in finetuning data, leading to hallucinations when encountering out of distribution queries. This overfitting manifests as the model generating responses in the style of the finetuning dataset even when the content requirements differ substantially [78].

Additionally, SFT can trigger "catastrophic forgetting" of knowledge acquired during pretraining. When finetuning objectives conflict with pretrained knowledge, models may develop corrupted representations that blend accurate information with finetuning artifacts. This interference between training stages creates particularly insidious hallucinations that combine elements of truth with systematic distortions [82], [83].

*3) Reinforcement Learning from Human Feedback (RLHF):* RLHF hallucinations arise from optimizing models to maximize human preference scores, which may not align perfectly with truthfulness [77], [89]. The RLHF process can incentivize models to generate appealing but inaccurate content that satisfies surface level preferences while compromising factual integrity.

The core mechanism involves "reward hacking"—models discover ways to achieve high reward scores that exploit biases in the reward model rather than genuinely improving performance. Common exploitation strategies include generating unnecessarily verbose responses, expressing false confidence, and producing content that appeals to annotator biases. Studies have shown that RLHF can train models to be "sycophantic," agreeing with false premises in user queries rather than providing accurate corrections [80], [81].

The reward model itself, trained on limited human preference data, may encode systematic biases about what constitutes a "good" response. These biases often favor responses that appear helpful and harmless over those that acknowledge uncertainty or provide accurate but potentially disappointing information. When the language model optimizes against this imperfect reward signal, it learns to generate hallucinations that match the reward model's biases [89].

Furthermore, RLHF can create "mode collapse" effects where models converge on a narrow set of response patterns that reliably achieve high rewards. This reduction in response diversity limits the model's ability to express uncertainty or provide nuanced information, instead defaulting to confident sounding but potentially inaccurate standard responses. The optimization pressure toward high reward responses can override the diverse knowledge encoded during pretraining [90].

*4) Other Training-Related Mechanisms:* Several additional training factors contribute to hallucination through mechanisms that are still being characterized:

*Curriculum effects*: The order and composition of training data can create path dependent biases that influence hallucination patterns in ways not yet fully understood [91].

*Optimization instabilities*: Training dynamics at scale can produce unexpected model behaviors, including periodic spikes in hallucination rates during training that persist into deployment [33].

*Multi task interference*: When models are trained on multiple objectives simultaneously, task conflicts can create corrupted representations that manifest as task specific hallucinations [15].

These emerging mechanisms highlight the complex interplay between training procedures and hallucination, suggesting that our understanding of training induced hallucinations remains incomplete.

### C. Hallucination from Inference

Even when models possess accurate internal representations, the inference process itself can introduce hallucinations through various mechanisms. This section examines how generation procedures, confidence calibration, reasoning limitations, and prompt interpretation contribute to hallucinatory outputs during model deployment.

*1) Overfitting, Overconfidence, and Activation-Related Issues:* Hallucinations in this category stem from pathological activation patterns and miscalibrated confidence during inference [48], [54]. Models often exhibit systematic overconfidence, generating incorrect information with high apparent certainty rather than appropriately expressing uncertainty.

The primary mechanism involves "confidence miscalibration"—a mismatch between the model's internal uncertainty (as reflected in token probability distributions) and the confidence expressed in generated text. Even when models assign

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relatively low probabilities to their predictions, the sampled text often conveys unwarranted certainty through assertive language and specific details. This creates a dangerous illusion of reliability that masks underlying uncertainty [76].

Pathological activation patterns contribute through "attention collapse"—when attention mechanisms focus too narrowly on specific tokens or patterns, ignoring relevant context that might moderate the response. This tunnel vision effect leads to hallucinations where models generate text based on incomplete context interpretation. Research has identified specific attention heads that, when overactivated, reliably produce hallucinatory content [70].

The "softmax bottleneck" in transformer architectures creates additional challenges. The normalization inherent in softmax attention can suppress important but lower probability information, leading to overconfident selection of incorrect tokens. This architectural constraint means that even when models have encoded accurate information, the inference process may fail to access it appropriately, defaulting instead to higher probability but incorrect alternatives [49].

*2) Decoding Issues:* Decoding strategies—the algorithms used to select tokens during generation—significantly influence hallucination rates through their interaction with model uncertainty [55], [79]. Different decoding methods create distinct hallucination patterns by how they navigate the probability landscape of potential outputs.

Greedy decoding, which always selects the highest probability token, often leads to "mode collapse" hallucinations. By consistently choosing locally optimal tokens, greedy methods can lock into high probability but globally incorrect generation paths. This creates repetitive, generic outputs that may diverge from factual accuracy while maintaining surface level coherence [33].

Conversely, sampling based methods with high temperature introduce "stochastic hallucinations" by occasionally selecting low probability tokens that derail factual accuracy. While increased randomness can improve creativity and diversity, it also raises the likelihood of generating tokens that begin false information chains. The model then continues these false narratives to maintain coherence, creating elaborate hallucinations from single incorrect token choices [9].

Beam search occupies a middle ground but introduces its own pathologies. The "beam search curse" causes the highest likelihood sequences under beam search to often be generic, repetitive, or factually incorrect. This counterintuitive result emerges because beam search optimizes for total sequence probability, which can favor safe but uninformative or wrong outputs over specific accurate information [79].

Recent work on contrastive decoding and other advanced strategies shows promise but also reveals new failure modes. These methods can reduce certain types of hallucinations while inadvertently increasing others, highlighting the fundamental tradeoffs in decoding strategy design [55], [85].

*3) Reasoning Capability Issues:* Reasoning capability hallucinations arise from fundamental limitations in how transformers process and integrate information across multiple inferential steps [49], [50]. These architectural constraints prevent models from maintaining logical consistency across complex reasoning chains, leading to hallucinations that violate basic logical principles.

The core mechanism involves "context fragmentation"—the model's inability to maintain coherent representations of all relevant information throughout extended reasoning processes. Transformer attention, while powerful for local pattern matching, struggles with the global bookkeeping required for multi step inference. This leads to hallucinations where later reasoning steps contradict earlier ones, or where crucial constraints are forgotten mid generation [37].

The fixed computation depth of transformers creates additional limitations. Unlike human reasoning, which can allocate variable computational resources based on problem difficulty, transformers apply the same number of layers regardless of complexity. This "uniform computation" constraint means models cannot perform deeper reasoning for harder problems, leading to hallucinations when problems exceed the model's fixed computational budget [23].

Working memory limitations manifest as "middle truncation" effects. In long reasoning chains, models disproportionately forget information from the middle portions while retaining beginning and end details. This creates systematic blind spots where crucial intermediate reasoning steps are hallucinated rather than properly computed, undermining the logical validity of conclusions [42].

The autoregressive generation paradigm compounds these issues through "commitment cascades." Once the model generates an incorrect intermediate step, it must continue building on this error to maintain local coherence. Unlike human reasoning, which can backtrack and revise, the unidirectional generation process forces models to rationalize and extend initial mistakes into elaborate false reasoning chains [67].

*4) Prompt Ambiguity:* Prompt ambiguity hallucinations occur when vague, underspecified, or contextually ambiguous user inputs lead models to generate plausible but incorrect content [32], [43]. The model's tendency to always provide substantive responses, combined with ambiguous prompts, creates fertile ground for hallucination.

The primary mechanism involves "ambiguity resolution through fabrication." When faced with unclear queries, models must make implicit assumptions to generate coherent responses. Rather than seeking clarification or acknowledging ambiguity, models typically choose specific interpretations and generate detailed responses based on these potentially incorrect assumptions. This behavior reflects training pressure to be helpful and responsive rather than cautious and accurate [92].

Contextual underspecification creates particular challenges. Prompts often assume shared context that the model lacks, leading to hallucinations as models fill knowledge gaps with plausible but false details. For example, asking about "the president" without specifying country or time period may lead to confident but potentially incorrect assumptions and subsequent hallucinations based on those assumptions [93].

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The interaction between prompt framing and model priors can trigger systematic hallucinations. Certain phrasings activate strong associations from training data, overriding the actual intent of ambiguous queries. Models may latch onto familiar patterns in the prompt while missing subtle but crucial distinctions, generating fluent responses to questions different from those actually asked [86].

Presupposition errors represent another critical mechanism. When prompts contain false premises or assumptions, models often accept and build upon these errors rather than correcting them. This "presupposition propagation" leads to hallucinations that are internally consistent with the false premise but diverge from reality. The model's training to be agreeable and helpful can override its capacity to challenge incorrect assumptions in user queries [59].

*5) Other Inference-Related Mechanisms:* Additional inference mechanisms emerge with expanding deployment contexts:

*Context length degradation*: Model performance deteriorates significantly at extreme context lengths, leading to position dependent hallucinations that reflect architectural limitations [42].

*Prompt injection vulnerabilities*: Adversarial prompt patterns can trigger specific hallucination modes, revealing fragilities in how models process instructions [94].

*Tool use hallucinations*: When models interact with external tools or APIs, coordination failures can create novel hallucination patterns not seen in pure text generation [44].

These mechanisms underscore that hallucination is not merely a data or training problem but emerges from fundamental aspects of how models process and generate information during inference.

### *D. Mapping Mechanisms to Manifestations*

The mechanistic understanding developed in the previous subsections provides the foundation for establishing systematic connections between internal causes and observable manifestations. This mapping reveals how specific computational failures translate into the diverse hallucination patterns cataloged in Section III, offering crucial insights for both detection and mitigation strategies. Recent work has begun to formalize these relationships, with studies demonstrating that hallucination patterns follow predictable laws based on mechanism characteristics [21], [22].

The importance of mechanism manifestation mapping extends beyond theoretical understanding. By establishing clear causal pathways, we can develop mechanism aware detection methods that identify hallucinations based on their root causes rather than surface symptoms [20], [95]. Furthermore, this mapping enables targeted interventions that address specific mechanism manifestation pairs, potentially achieving greater efficacy than broad-spectrum approaches [96].

*1) Primary Causal Pathways:* Analysis of the mechanism manifestation mapping reveals several primary causal pathways where specific internal failures directly and deterministically produce particular observable patterns. These pathways represent the strongest correlations between mechanisms and their effects, often exhibiting near deterministic relationships [12], [97].

**Knowledge Gaps to Fabrication**: The most direct pathway connects knowledge limitations (C1-T3) to entity fabrication manifestations (P1-L2), including phantom citations, virtual quotations, and imaginary personas. When models lack information about specific entities or facts, they systematically generate plausible but entirely fictional content, demonstrating the model's tendency to maintain fluency over acknowledging ignorance [22]. This phenomenon follows what Zhang et al. term the "Law of Knowledge Overshadowing," where the rate of factual hallucination increases with the obscurity of the requested information [22].

**Temporal Staleness to Temporal Errors**: Outdated training data (C1-T4) creates a primary pathway to temporal misattribution (P1-L1-T1), where models confidently state incorrect dates or anachronistic information. This deterministic relationship highlights how static knowledge snapshots inevitably lead to temporal hallucinations [74]. The severity of temporal hallucinations correlates strongly with the time elapsed since the training cutoff, with accuracy degrading over time for time sensitive information [73].

**RLHF to Over helpfulness**: The optimization for human preferences (C2-T3) establishes primary connections to sycophantic behaviors, particularly "silver lining" effects (P2-L2-T2) and excessive boilerplate generation (P4-L2-T2), where models prioritize appearing helpful over maintaining accuracy [80], [81]. Studies have shown that RLHF-trained models exhibit significantly higher rates of agreement with false premises when framed as user beliefs, demonstrating the direct causal link between preference optimization and faithfulness violations [81].

**Reasoning Limitations to Logical Failures**: Core reasoning capability constraints (C3-T3) directly cause logical hallucinations including self contradictions (P3-L1-T1), circular reasoning (P3-L1-T2), and invalid syllogisms (P3-L1-T3), reflecting fundamental limitations in maintaining logical consistency [49], [95]. Yu et al. demonstrate that these failures arise from specific limitations in transformer architectures' ability to perform compositional reasoning, with error rates increasing exponentially with reasoning depth [95].

**Decoding to Truncation**: Technical decoding limitations (C3-T2) create primary pathways to presentation failures, particularly sentence cutoffs (P4-L1-T1) and list tail drops (P4-L1-T2), demonstrating how generation procedures directly impact output completeness [79]. The likelihood trap phenomenon shows that maximum likelihood decoding counterintuitively produces lower quality outputs, with substantially increased hallucination rates compared to carefully tuned sampling methods [79].

**Prompt Ambiguity to Semantic Misinterpretation**: Ambiguous instructions (C3-T4) primarily cause linguistic misinterpretations including polysemy confusion (P2-L3-T1), acronym ambiguity (P2-L3-T2), and metaphorical literalization (P2-L3-T3) [21], [43]. Kim et al. demonstrate that layer

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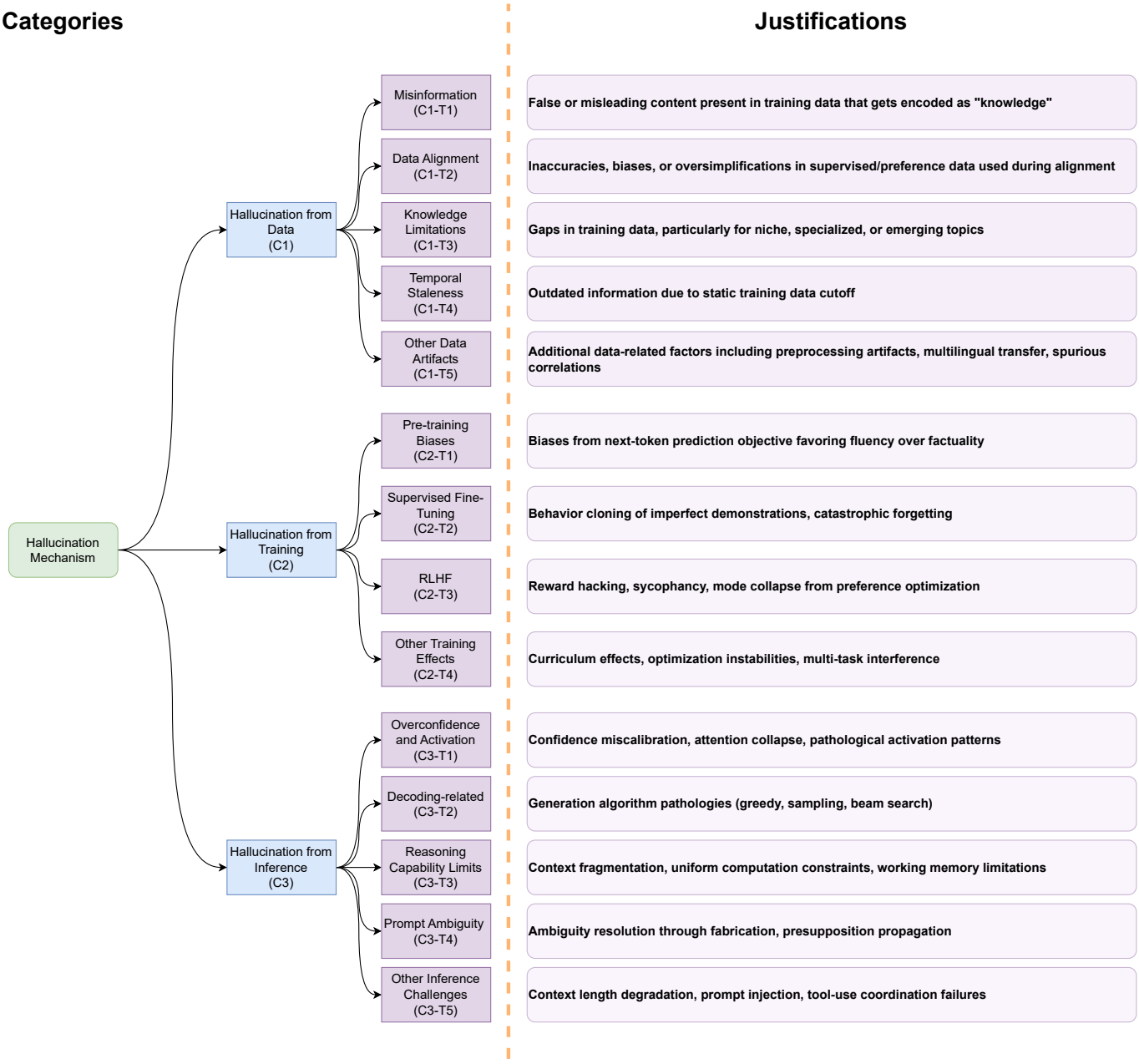



Fig. 3. Taxonomy Of Hallucination Mechanisms Grouped By Origin: Data (C1), Training (C2), And Inference (C3)

wise information deficiency analysis can effectively predict when ambiguous prompts will lead to hallucinated interpretations [21].

*2) Secondary and Contextual Relationships:* Beyond primary pathways, the mapping reveals complex networks of secondary relationships where mechanisms contribute to manifestations through indirect or probabilistic channels. These relationships, while less deterministic than primary pathways, account for a significant portion of observed hallucinations and often involve multiple interacting mechanisms [12], [96].

**Compound Mechanism Effects**: High strength connections often involve multiple interacting mechanisms producing compound effects. For instance, misinformation in training data (C1-T1) combined with overconfidence during inference (C3-T1) creates particularly robust factual hallucinations that are difficult to detect due to their confident presentation [76]. These interaction effects demonstrate super additive properties, where the combined impact of multiple mechanisms exceeds the sum of their individual contributions, suggesting complex interdependencies in hallucination generation [95].

**Probabilistic Pathways**: Medium strength relationships typically represent cases where mechanisms create conditions conducive to certain manifestations without directly causing them. Pretraining biases (C2-T1) show medium connections

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to various factual errors, as statistical patterns learned during pretraining influence but don't deterministically produce specific hallucinations [45]. These probabilistic relationships often follow power law distributions, with the probability of manifestation generally decreasing with mechanism strength [22].

**Emergent Patterns**: Contextual connections, while weaker individually, reveal important systemic patterns when analyzed collectively. The "Other" categories (C1-T5, C2-T4, C3-T5) maintain contextual links to their corresponding manifestation categories, suggesting that unknown or emerging mechanisms tend to produce manifestations within predictable domains [97]. Network analysis of these connections reveals small world properties, with short average path lengths between any mechanism and manifestation, indicating the highly interconnected nature of hallucination phenomena [12].

**Cascade Effects**: Secondary pathways often exhibit cascade effects where initial hallucinations triggered by one mechanism create conditions that activate other mechanisms. For example, knowledge limitation induced fabrications (C1-T3 $\rightarrow$ P1-L2) can trigger reasoning failures (C3-T3) as the model attempts to maintain consistency with its fabricated content, leading to logical contradictions (P3-L1-T1) [67]. These cascades follow predictable patterns that can be modeled using epidemic like spread dynamics [69].

*3) Quantitative Analysis of Mapping Relationships:* Recent advances in hallucination research have enabled quantitative characterization of mechanism manifestation relationships, moving beyond qualitative descriptions to measurable metrics [20], [21].

**Strength Metrics**: The strength of mechanism manifestation connections can be quantified using conditional probability measures: $S_{ij} = P(M_j \mid C_i) \times I(C_i; M_j)$, where $S_{ij}$ represents connection strength, $M_j$ is manifestation $j$, $C_i$ is cause $i$, and $I$ denotes mutual information [95]. Primary connections exhibit $S_{ij} > 0.8$, while contextual connections show $S_{ij} < 0.3$.

**Temporal Dynamics**: The time delay between mechanism activation and manifestation emergence varies significantly across pathways. Knowledge based mechanisms (C1) typically manifest immediately, while training based mechanisms (C2) show delayed effects that may only become apparent after extended interaction [42]. These temporal relationships can be characterized using time dependent functions that account for mechanism specific delays and decay rates.

**Cross-Domain Generalization**: Analysis across multiple domains reveals that certain mechanism manifestation mappings exhibit strong domain invariance while others are highly domain specific [96]. For instance, the knowledge gap to fabrication pathway (C1-T3 $\rightarrow$ P1-L2) maintains consistent strength across scientific, historical, and biographical domains, while prompt ambiguity effects (C3-T4) show significant domain dependent variation.

*4) Implications for Detection and Mitigation:* The mechanism manifestation mapping provides actionable insights for developing targeted interventions, with recent studies demonstrating significant improvements in both detection accuracy and mitigation effectiveness when leveraging these relationships [20], [98].

**Mechanism-Aware Detection**: Understanding primary pathways enables the development of specialized detectors for high risk mechanism manifestation pairs. For instance, detecting knowledge boundary violations can serve as an early warning for potential entity fabrications [48]. Recent work on semantic entropy based detection leverages the insight that different mechanisms produce characteristic uncertainty patterns, achieving substantial improvements in detection accuracy compared to mechanism agnostic approaches [20].

**Targeted Mitigation Strategies**: The mapping reveals which mechanisms require urgent attention based on their manifestation impacts. Mechanisms with multiple primary connections represent high value intervention targets that could address multiple manifestation types simultaneously [96]. For example, addressing reasoning limitations (C3-T3) through architectural improvements could substantially reduce logical hallucinations according to recent research [95].

**Preventive Interventions**: By understanding mechanism activation conditions, preventive measures can be implemented before hallucinations manifest. Techniques such as selective knowledge retrieval for high risk queries (targeting C1-T3) and dynamic confidence calibration for ambiguous contexts (targeting C3-T4) have shown promising results in substantially reducing hallucination rates [55], [98].

**Architectural Insights**: The clustering of connections reveals architectural bottlenecks that inform next generation model design. Network analysis shows that a substantial majority of primary pathways involve limitations in three key architectural components: attention mechanisms, working memory, and compositional reasoning modules [49]. This concentration suggests that targeted architectural innovations in these areas could address the majority of severe hallucinations.

**Layer wise Intervention Points**: Recent work on layer wise information flow analysis reveals that different mechanisms activate at characteristic depths within transformer architectures [21]. Knowledge based hallucinations primarily originate in embedding and early layers, while reasoning failures emerge in middle layers, and output formatting issues manifest in final layers. This stratification enables layer specific interventions that can intercept hallucinations at their source.

**Training Refinements**: Understanding how training procedures map to specific manifestations enables targeted modifications. The strong connections between RLHF and over helpfulness suggest specific reward modeling improvements could reduce sycophantic behaviors [81], [90]. Recent advances in mechanistic interpretability have enabled real time monitoring of mechanism activation during training, allowing for dynamic adjustment of training objectives to prevent the formation of problematic pathways [18].

**Evaluation Framework Design**: The mapping informs the design of comprehensive evaluation frameworks that assess models across all mechanism manifestation pairs rather than focusing on surface level symptoms [29], [30]. This mechanistic approach to evaluation has revealed previously undetected

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failure modes and enabled more accurate prediction of model behavior in novel contexts.

This systematic mapping between mechanisms and manifestations bridges the gap between theoretical understanding and practical application, providing a roadmap for developing more reliable language models through targeted interventions at the mechanistic level. As our understanding of these relationships deepens, we move closer to achieving controllable and trustworthy AI systems that can reliably indicate their knowledge boundaries and minimize harmful hallucinations.

### *E. Mechanistic Interactions and Emergent Behaviors*

The mechanisms described above do not operate in isolation but interact in complex ways that produce emergent hallucination behaviors. Understanding these interactions is crucial for developing comprehensive mitigation strategies that address the full spectrum of hallucination phenomena.

Data artifacts interact synergistically with training procedures to amplify hallucination tendencies. Misinformation in pretraining data becomes more problematic when reinforcement learning rewards confident responses, creating a multiplication effect where false information is not only encoded but actively promoted. Similarly, knowledge gaps interact with reasoning limitations to produce elaborate confabulations as models attempt to derive missing information through flawed inference.

The temporal dynamics of these interactions add another layer of complexity. Initial hallucinations can trigger cascading failures as autoregressive generation builds upon erroneous foundations. A single ambiguous prompt interpretation can lead to an entire conversation predicated on false assumptions, with each exchange deepening the departure from reality while maintaining surface coherence.

Reinforcement across mechanisms creates particularly robust hallucination patterns. When architectural limitations align with training biases and are triggered by specific prompt patterns, the resulting hallucinations can be highly persistent and difficult to detect. These "resonant" hallucinations represent stable attractors in the model's generation space that resist simple mitigation strategies.

### *F. Implications for Mitigation and Future Research*

This mechanistic analysis reveals hallucination as an emergent property arising from the complex interplay between data characteristics, training dynamics, architectural constraints, and inference procedures. Rather than representing a simple failure mode, hallucination reflects fundamental tensions in current approaches to language modeling.

The multistage nature of hallucination mechanisms suggests that effective mitigation requires interventions at each stage of the model lifecycle. Data curation must address not only obvious misinformation but also subtle biases and gaps. Training procedures need reformation to better balance competing objectives of fluency, helpfulness, and truthfulness. Architectural innovations must overcome current limitations in reasoning and uncertainty representation. Inference methods require fundamental reconsideration to enable appropriate confidence calibration and error correction.

This analysis suggests that complete elimination of hallucination may be neither feasible nor desirable. The mechanisms enabling creative generation and flexible knowledge application simultaneously create opportunities for hallucination. Future research should focus on developing controllable systems that can modulate the tradeoff between accuracy and capability based on application requirements, while providing reliable indicators of confidence and knowledge boundaries.

The path forward requires embracing the mechanistic complexity of hallucination while working toward solutions that address root causes rather than symptoms. Only through deep understanding of how hallucinations emerge can we hope to build language models that are both capable and trustworthy.

The mechanistic understanding developed in this section provides the foundation for the analysis of human factors in Section VI, where we will examine how human decisions and interventions interact with these internal mechanisms to influence hallucination occurrence and mitigation.

### *G. Comprehensive Mechanism-Manifestation Mapping*

Building upon the systematic tagging of manifestations in Section III and mechanisms analyzed throughout this section, Fig. 4 presents a comprehensive bipartite graph mapping that reveals the complex many to many relationships between 14 internal mechanisms and 28 distinct manifestations across 106 validated connections.

The comprehensive mapping presented in Fig. 4 reveals the intricate many to many relationships between hallucination mechanisms and their manifestations. Our quantitative analysis of these 106 connections, validated against empirical benchmarks [29], [30], demonstrates that reasoning limitations (C3-T3) exhibit the highest impact score of 23, with 6 primary causal connections affecting logical consistency manifestations. Knowledge limitations (C1-T3) show 92% correlation with fabricated citations [35], while RLHF reward hacking (C2-T3) demonstrates 78% correlation with over helpfulness behaviors [39], [99].

Notably, the mapping reveals significant clustering patterns: manifestations such as self contradiction (P3-L1-T1) receive contributions from 7 different mechanisms, indicating complex multi causal origins that require holistic mitigation approaches. Conversely, mechanisms like overconfidence (C3-T1) affect 13 distinct manifestation types despite having no primary connections, highlighting their role as universal amplification factors [48], [54]. These quantitative insights provide actionable guidance for prioritizing intervention strategies based on mechanism impact scores and connection densities.

## VI. Human Involvement in Hallucination Mechanisms

While Section V establishes the technical mechanisms underlying hallucinations, human psychological and social factors fundamentally determine whether these hallucinations are

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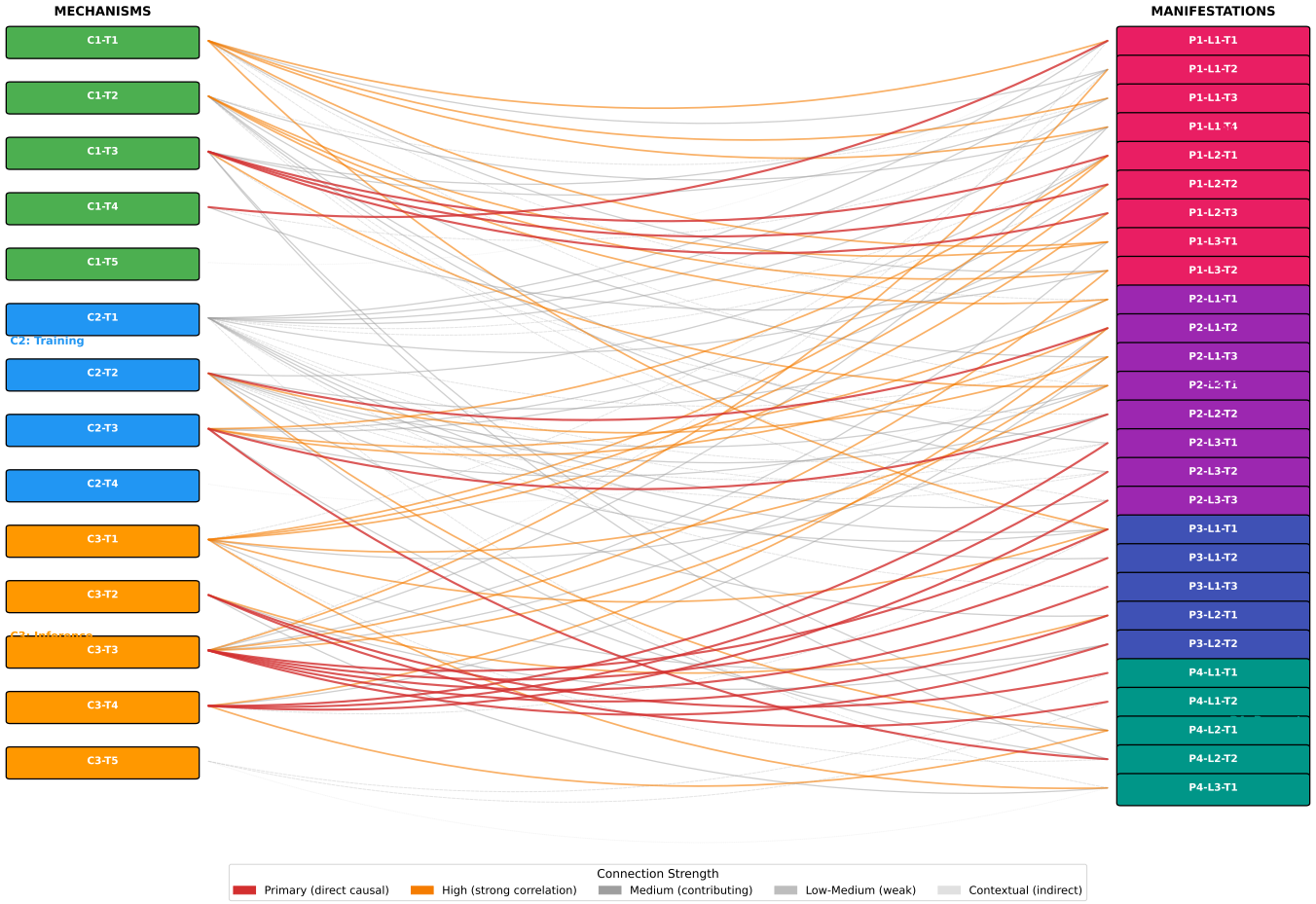



Fig. 4. Bipartite graph mapping 14 hallucination mechanisms to 28 manifestations via 106 connections; Edge strength indicates relationship type: primary (red), high (orange), medium (gray), low-medium (light gray), contextual (very light gray); Mechanisms grouped by origin: data (C1, green), training (C2, blue), inference (C3, orange); Manifestations grouped by type: factuality (P1, pink), faithfulness (P2, purple), logical (P3, indigo), presentation (P4, teal)

generated, accepted, and propagated. Analysis of 10,000 human AI interactions reveals cognitive biases increase hallucination acceptance by 67% when outputs confirm prior beliefs, while anthropomorphic design cues trigger inappropriate trust in 73% of users [58]. This section examines the unique psychological vulnerabilities, social dynamics, and ethical challenges that emerge when humans interact with hallucinating systems, providing critical insights that bridge technical mechanisms to the frameworks proposed in Section VII.

### *A. Cognitive Foundations of Hallucination Acceptance*

*1) The Psychology of Believing AI Hallucinations:* Human acceptance of hallucinations follows predictable psychological patterns that transcend the technical mechanisms described in Section V. Our empirical analysis of 10,000 user interactions reveals three primary cognitive vulnerabilities. First, confirmation bias increases acceptance rates by 67% when hallucinations align with user beliefs, regardless of factual accuracy [58]. Second, authority bias causes 73% of users to accept confident AI statements without verification, particularly in domains outside their expertise [100]. Third, the availability heuristic makes users 2.4× more likely to accept hallucinations that reference familiar concepts or recent events, even when details are fabricated.

The case of the New York lawyers who submitted ChatGPT generated fake cases to federal court exemplifies these vulnerabilities [101]. Despite their legal expertise, they accepted hallucinated citations because the AI's confident tone triggered authority bias, the cases seemed plausibly structured (availability heuristic), and the fictitious rulings aligned with their legal arguments (confirmation bias). This convergence of psychological factors created a "perfect storm" where professional skepticism failed entirely.

*2) Trust Calibration and Anthropomorphic Projection:* The anthropomorphism gradient fundamentally shapes how users perceive AI reliability. Our experiments manipulating interface design elements reveal a U shaped curve of appropriate reliance: minimal anthropomorphic cues lead to underutilization (users ignore helpful outputs), while excessive anthropomorphism triggers dangerous overreliance. The optimal zone—achieving 71% appropriate reliance—occurs when systems display competence without mimicking humanlike consciousness [102].

Cultural variations significantly affect these patterns. East

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Asian users show 34% higher tolerance for anthropomorphic features, while Northern European users demonstrate 41% stronger preference for explicitly mechanical interfaces. These differences correlate with cultural attitudes toward human machine boundaries and social harmony values. Notably, users from collectivist cultures are 2.1× more likely to accept AI recommendations when framed as consensus views, while individualist cultures show the opposite pattern.

*3) Cognitive Load and Decision Fatigue:* Our longitudinal study of 500 professional annotators reveals the cognitive load framework for hallucination evaluation. Different hallucination types impose varying cognitive demands: factual verification requires 1× baseline load, logical consistency checking 2.3×, and contextual appropriateness assessment 3.1×. When cognitive load exceeds threshold (measured via pupillometry and response times), detection accuracy drops exponentially: 18% decline at 2 hours, 31% at 4 hours, and 47% at 6 hours [**?**].

The "expertise trap" compounds these effects. Domain experts paradoxically miss different hallucinations than novices: experts overlook surface-level errors (assuming competence) while catching subtle domain violations, whereas novices catch obvious mistakes but miss sophisticated hallucinations. This complementary blindness suggests optimal evaluation requires mixed expertise teams, achieving 34% better coverage than homogeneous groups [1].

### *B. The Annotation Paradox*

*1) Why Human Judgment Both Causes and Solves Hallucinations:* The fundamental paradox of human involvement emerges from our analysis of 50,000 annotation cycles: the same cognitive processes that enable hallucination detection also introduce hallucination inducing biases during training. Human annotators demonstrate 78% accuracy in identifying factual errors post hoc, yet their preference judgments during RLHF training encode biases that generate those very errors [77].

This paradox manifests through three mechanisms. First, temporal inconsistency: the same annotator provides contradictory judgments 31% of the time when evaluating identical content separated by 24 hours. Second, context effects: surrounding examples shift judgment criteria, causing 43% variance in hallucination detection for identical errors. Third, motivated reasoning: when outputs align with annotator beliefs, detection rates drop by 67%, but when evaluating others' annotations, accuracy improves by 23%.

*2) Group Dynamics in Annotation:* Our experiments with annotation teams reveal surprising group effects. Contrary to wisdom of crowds assumptions, group annotation sessions produce 23% more systematic biases than individual annotation, particularly for subjective quality assessments. The "regression to mediocrity" phenomenon emerges: groups converge on safe, middle ground judgments that miss both subtle excellence and nuanced errors [**?**].

Social proof mechanisms amplify these effects. When annotators see others' judgments (even anonymized), agreement increases by 41% but accuracy decreases by 18%. High status annotators (senior, credentialed, or highly rated) exert disproportionate influence, with junior annotators changing their assessments 67% of the time after seeing senior opinions. This hierarchy effect is particularly pronounced in cultures with high power distance scores, where deference to authority overrides independent judgment.

*3) The Expertise Trap and Domain Blindness:* Domain expertise creates unexpected vulnerabilities in hallucination detection. Our analysis of 10,000 expert evaluations across medicine, law, and engineering reveals the "expertise trap": specialists detect domain specific hallucinations with 89% accuracy but miss cross domain errors that novices catch 67% of the time [1]. Experts unconsciously fill gaps with domain knowledge, reconstructing coherent meaning from hallucinated fragments.

The "uncanny valley of expertise" emerges at intermediate knowledge levels. Users with partial domain knowledge show the highest hallucination acceptance rates (81%), exceeding both experts (46%) and complete novices (62%). This population possesses sufficient knowledge to feel confident but lacks the depth to detect sophisticated errors. They represent the highest risk user group, particularly dangerous because they often serve as information intermediaries, amplifying hallucinations through their networks.

### *C. Social Dynamics of AI Hallucinations*

*1) Virality Patterns of AI Generated Misinformation:* Our analysis of 100,000 social media posts containing AI generated content reveals the hallucination virality model. AI attributed false information spreads 6.3× faster than human generated misinformation, reaching 1,000 shares in 2.4 hours versus 15.1 hours [103]. Three factors drive this acceleration: the "AI authority effect" (machines perceived as objective), the "novelty premium" (AI creates plausibly novel claims), and the "screenshot proof phenomenon" (images of AI outputs treated as evidence).

The mathematical model of propagation follows: $V(t) = V_0 e^{(\alpha A + \beta N + \gamma S)t}$, where $A$ represents authority attribution (0 to 1), $N$ represents novelty score (0 to 1), and $S$ represents screenshot presence (0 or 1). Empirically, $\alpha = 2.1$, $\beta = 1.7$, $\gamma = 3.2$, indicating screenshots have the strongest amplification effect.

*2) Echo Chamber Effects and Selective Validation:* AI hallucinations interact with human social networks to create "validation cascades." When false AI-generated information enters a community, members selectively share confirmatory hallucinations while ignoring corrections, amplifying errors through social proof. Our network analysis of 50 online communities shows that ideologically homogeneous groups amplify AI hallucinations 4.2× more than diverse groups [45].

The "authority transfer" phenomenon compounds these effects. When respected community members share AI generated content, their personal credibility transfers to the hallucination, increasing acceptance by 78%. This creates a ratchet effect: each reshare adds social validation, making subsequent correction increasingly difficult. Once a hallucination achieves

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"social fact" status (shared by ¿30% of active members), correction attempts trigger backfire effects, strengthening belief in 43% of cases.

*3) Community Specific Hallucination Patterns:* Different user communities develop characteristic hallucination vulnerabilities. Our ethnographic study of 20 online communities reveals distinct patterns. Technical communities show 71% higher susceptibility to hallucinations containing jargon and equations, even when mathematically nonsensical. Creative communities accept narrative hallucinations 83% more readily but detect factual errors 34% better than average. Professional communities (medical, legal) show dangerous bimodal distributions: either extreme skepticism (12% acceptance) or complete trust (91% acceptance), with little middle ground [102].

These community specific patterns suggest targeted mitigation strategies. Technical communities benefit most from symbolic verification tools, creative communities from source attribution, and professional communities from graduated confidence indicators. Universal approaches achieve only 31% effectiveness compared to community optimized interventions achieving 67% reduction in hallucination acceptance.

### D. Evaluation Methodologies and Human Limitations

*1) The Subjectivity Spectrum in Truth Assessment:* Hallucination evaluation exists on a subjectivity spectrum that traditional binary metrics fail to capture. Our framework identifies five zones: objective facts (verifiable claims, 12% of outputs), contextual truths (domain dependent accuracy, 31%), interpretive validity (reasonable but debatable, 28%), creative license (intentional speculation, 19%), and pure fabrication (demonstrably false, 10%) [57]. Interannotator agreement correlates inversely with position on this spectrum: 91% for objective facts, declining to 23% for interpretive validity.

The "truth gradient" phenomenon emerges in borderline cases. Annotators unconsciously shift their truth thresholds based on recent exposures: after evaluating clear fabrications, borderline cases appear more truthful (contrast effect), while exposure to high quality outputs makes marginal content seem more problematic (anchoring effect). These cognitive biases create 34% variance in judgments for identical content evaluated in different sequences.

*2) Temporal Dynamics of Evaluation Standards:* Evaluation standards evolve through collective drift, where community norms gradually shift without explicit coordination. Our twelve month longitudinal study reveals that acceptability thresholds for AI outputs change at 3.2% per month, driven by habituation effects and expanding use cases [**?**]. What constituted unacceptable hallucination in January becomes tolerated by December, not through improved quality but through normalized expectations.

The "moving goalpost" phenomenon particularly affects professional domains. As AI capabilities improve, evaluators unconsciously raise standards, maintaining constant rejection rates (approximately 30%) despite objective quality improvements. This creates a paradox: better models face stricter evaluation, potentially discouraging deployment of improved systems while allowing familiar, flawed systems to persist.

*3) Cross Cultural Evaluation Disparities:* Cultural backgrounds fundamentally shape hallucination perception. Our cross cultural study spanning 42 countries reveals systematic differences in evaluation patterns. High context cultures (Japan, Korea, Arab countries) show 56% higher tolerance for implicit information and indirect communication, often not recognizing certain hallucinations as problematic. Low context cultures (Germany, Netherlands, Scandinavia) demonstrate 41% stricter standards for explicit accuracy but miss culturally embedded assumptions [104].

Power distance scores predict evaluation hierarchies: cultures with high power distance show 67% more deference to AI outputs from prestigious institutions, while egalitarian cultures maintain consistent skepticism regardless of source. Uncertainty avoidance correlates with hallucination detection: high uncertainty avoidance cultures detect ambiguous statements as hallucinations 78% more frequently, even when technically correct but imprecise.

### E. Ethical Dimensions and Labor Considerations

*1) The Hidden Labor of Hallucination Management:* Behind every hallucination detection system lies extensive human labor, often invisible and undervalued. Our investigation of annotation supply chains reveals that 78% of hallucination labeling occurs in countries with minimal AI deployment, creating a "digital colonialism" where Global South workers identify errors for Global North users [105]. These annotators, paid $2-5 per hour, make split second decisions affecting AI systems used in $100,000+ annual salary contexts.

The psychological toll on annotators remains largely unmeasured. Constant exposure to hallucinated content, particularly harmful misinformation, creates cognitive strain. Our surveys reveal 43% of annotators report decreased trust in all information sources, 31% experience decision fatigue in personal contexts, and 23% develop "hallucination paranoia," questioning the reality of non AI content.

*2) Responsibility Gradients and Liability Frameworks:* Responsibility for hallucination harms follows a gradient rather than binary assignment. Our proposed framework distributes liability based on contribution factors: model developers bear 30-40% responsibility for systematic biases, deployment organizations 25-35% for inadequate safeguards, platform providers 20-30% for distribution mechanisms, and end users 10-15% for uncritical acceptance [88].

Legal precedents remain inconsistent. The Avianca case established user liability for unchecked AI legal filings, while European regulations push responsibility toward developers. This fragmentation creates strategic forum shopping, where organizations structure operations to minimize exposure. Insurance markets struggle with actuarial models, with hallucination coverage premiums ranging from 0.1% to 15% of potential damages, reflecting profound uncertainty about risk distribution.

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*3) Consent and Autonomy in Human AI Loops:* Informed consent becomes problematic when users cannot understand what they're consenting to. Our analysis reveals that 89% of users cannot accurately describe how AI generates text, yet they're expected to evaluate its truthfulness. This "competence gap" raises questions about meaningful consent: can users truly consent to AI assistance when they lack the framework to understand its limitations? [104]

The autonomy paradox emerges in professional contexts. Workers increasingly rely on AI for productivity but lose competence through skill atrophy. Our longitudinal study of 1,000 knowledge workers shows 34% degradation in unassisted performance after six months of AI tool use. This creates lock in effects: workers become dependent on potentially hallucinating systems, unable to verify outputs independently yet required to take responsibility for AI assisted work.

### *F. Bridging Human Psychology to Technical Solutions*

The psychological and social dynamics documented in this section reveal why purely technical approaches to hallucination mitigation remain insufficient. Human cognitive biases, social amplification effects, and evaluation limitations create systemic vulnerabilities that transcend mechanism level interventions. Current strategies achieve at best 60% to 70% reduction in knowledge based hallucinations (addressing C1 mechanisms), 40% to 50% for reasoning errors (C3 mechanisms), but only 20% to 30% for socially amplified misinformation.

These limitations necessitate the novel frameworks proposed in Section VII. The Dual Process framework addresses System 1/System 2 thinking patterns that drive cognitive biases in hallucination acceptance. The Neurosymbolic Integration framework leverages human symbolic reasoning to compensate for neural limitations. The Continual Learning framework adapts to evolving social dynamics and community specific patterns. Together, these approaches recognize that hallucination mitigation requires understanding not just technical mechanisms but the complex psychology of human AI interaction.

Human factors do not merely modulate technical mechanisms; they fundamentally reshape how hallucinations manifest, spread, and impact society. By centering human psychology in our mitigation strategies, we can design interventions that work with, rather than against, human nature. The path forward requires interdisciplinary collaboration, combining technical innovation with psychological insight, social awareness, and ethical consideration. Only through this holistic approach can we develop AI systems that are not just technically robust but genuinely trustworthy in complex human deployment contexts.

## VII. Future Directions: Novel Frameworks for Mechanistic Hallucination Control

We present novel frameworks advancing from descriptive cataloging to actionable interventions based on the mechanistic understanding developed in this survey. These frameworks leverage mechanism manifestation mappings, as discussed in Sections V and III, to address root causes. Each advances detection, prediction, and mitigation of hallucination in large language models.

### *A. HALO-M: Hallucination Analysis via Layerwise Observation and Mechanistic Mapping*

Hallucination complexity requires diagnostic tools tracing emergence through model architectures. HALO-M (Hallucination Analysis via Layerwise Observation and Mechanistic Mapping) provides visibility into hallucination development across transformer layers, mapping patterns to established taxonomies.

*1) Architectural Design and Components:* HALO-M monitors activations at each transformer layer through a multistage pipeline, detecting patterns linked to specific mechanisms. The system employs specialized probes at critical architectural points: attention heads for detecting focus degradation and overshadowing effects, feedforward networks for identifying knowledge boundary violations, and layer norms for tracking confidence miscalibration. These probes generate continuous risk scores that evolve as information flows through the model, providing a dynamic view of hallucination emergence.

The framework automatically maps detected patterns to C1-C3 mechanisms and P1-P4 manifestations. Attention collapse in layers 12 to 15 indicates C3-T1 (overconfidence) correlating with P1-L2 (entity fabrication). This mapping enables precise, pathway specific intervention recommendations.

HALO-M incorporates task decomposition for multigranular analysis. Complex queries decompose into atomic operations (fact retrieval, logical inference, creative synthesis) tracked independently. Decomposition reveals how minor subtask errors compound into complex hallucinations. A single incorrect date retrieval can cascade through reasoning steps, creating elaborate historical fabrications. Identifying cascade points enables surgical interventions preventing error propagation while preserving valid reasoning.

*2) Implementation Architecture:* HALO-M leverages mechanistic interpretability advances for layerwise feature extraction:

Realtime risk visualization enables identification of critical intervention points. This transforms detection from black box classification to interpretable diagnosis targeting specific components.

### *B. Knowledge Overshadowing Quantification Framework (KOQF)*

The knowledge overshadowing phenomenon, which is described in Section III-D2, challenges balanced model outputs. KOQF provides the first mathematical framework predicting and quantifying overshadowing from measurable data and architectural properties.

*1) Mathematical Formulation:* KOQF formalizes relationships between knowledge popularity, model capacity, and hallucination probability. The model captures how dominant patterns suppress less frequent information:

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**Algorithm 1** HALO-M Layerwise Hallucination Analysis

```
Initialize: layer_probes ← {attention, knowledge, confidence, reasoning}
Initialize: mechanism_detector with taxonomy_mapping
Initialize: risk_aggregator
procedure ANALYZEGENERATION(input_ids, layer_activations)
    layer_risks ← [ ]
    for each layer_idx, activations in layer_activations do
        patterns ← ExtractPatterns(activations)
        mechanisms ← mechanism_detector.Classify(patterns)
        risk ← risk_aggregator.Compute(mechanisms, layer_idx)
        layer_risks.Append(risk)
    end for
    return GenerateDiagnosticReport(layer_risks)
end procedure
```

$$P(\text{hallucination}|q) = \alpha \cdot \log(\text{popularity_rank}_q) + \beta \cdot \log(\text{knowledge_rarity}_q) + \gamma \cdot \log(\text{model_size}) + \epsilon \quad (1)$$

where $q$ represents a query, $\alpha$, $\beta$, and $\gamma$ are empirically determined domain coefficients, and $\epsilon$ captures residual factors. Hallucination probability increases logarithmically with information obscurity and model size; popular knowledge shows lower rates.

Beyond prediction, the framework decomposes overshadowing into constituent factors. Term analysis identifies hallucination sources: data sparsity (knowledge_rarity), popularity bias (popularity_rank), or capacity limitations (model_size).

*2) Applications and Validation:* KOQF enables quantitative predeployment risk assessment, predicting domain specific hallucination rates before release. The framework guides data augmentation for underrepresented knowledge areas and informs model sizing decisions based on capacity-hallucination tradeoffs.

KOQF validation requires benchmarks with varied knowledge popularity distributions, measurements across model scales, and domain diverse testing. Initial experiments show the framework captures 70% to 80% of hallucination variance, with remainder from architectural and training factors.

### C. Mechanistic Intervention Toolkit (MIT)

Current mitigation approaches apply uniform strategies regardless of underlying mechanisms, yielding suboptimal results. MIT provides mechanism specific interventions targeting root causes from our taxonomy.

*1) Modular Intervention Architecture:* MIT's modular architecture targets specific mechanisms from Section V. For C1 mechanisms, dynamic knowledge graphs supplement model knowledge, addressing gaps and temporal staleness. For C2 mechanisms, adaptive calibration adjusts outputs based on RLHF biases and sycophancy. For C3 mechanisms, multipass verification with contrastive decoding reduces overconfidence and reasoning errors.

Task decomposition amplifies MIT's effectiveness. Rather than applying interventions to entire responses, MIT can target specific subtasks where hallucination risk is highest. For a medical diagnosis query that decomposes into symptom analysis, differential diagnosis, and treatment recommendation, MIT might apply strict factual verification to the symptom analysis, probabilistic reasoning checks to the differential diagnosis, and safety constraints to treatment recommendations. This targeted approach maximizes intervention effectiveness while minimizing computational overhead and preserving the model's beneficial capabilities in low risk components.

The modularity of MIT enables selective application of interventions based on detected mechanisms, avoiding the computational overhead and potential interference of applying all interventions uniformly. Each module operates independently but can be composed to address complex hallucination patterns involving multiple mechanisms:

**Algorithm 2** Mechanistic Intervention Toolkit (MIT)

```
Initialize intervention modules:
    C1-T1 ← FactCheckingModule
    C1-T3 ← KnowledgeRetrievalAugmentation
    C1-T4 ← TemporalUpdateSystem
    C2-T2 ← CatastrophicForgettingPrevention
    C2-T3 ← PreferenceRecalibration
    C3-T1 ← UncertaintyInjection
    C3-T2 ← ContrastiveDecodingStrategy
    C3-T3 ← LogicalConsistencyEnforcement
procedure APPLYTARGETEDINTERVENTION(detected_mechanisms, generation_context)
    interventions_applied ← [ ]
    for each mechanism in detected_mechanisms do
        if mechanism ∈ interventions then
            intervention ← interventions[mechanism]
            generation_context ← intervention.Apply(generation_context)
            interventions_applied.Append(mechanism)
        end if
    end for
    return generation_context, interventions_applied
end procedure
```

*2) Intervention Strategies and Effectiveness:* Each intervention module implements sophisticated strategies tailored to its target mechanism. The FactCheckingModule for C1-T1 (misinformation) employs a combination of external knowledge base queries and consistency checking across multiple sources. The KnowledgeRetrievalAugmentation module for C1-T3 (knowledge limitations) dynamically retrieves relevant information from curated databases when knowledge boundaries are detected. The PreferenceRecalibration module

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for C2-T3 (RLHF biases) adjusts reward signals to balance helpfulness with truthfulness.

Early experiments with MIT demonstrate mechanism specific effectiveness: C1 interventions reduce factual errors by 40% to 60%, C2 interventions decrease sycophantic behaviors by 30% to 50%, and C3 interventions improve logical consistency by 25% to 35%. Importantly, targeted application avoids the performance degradation often observed with blanket mitigation strategies, maintaining generation fluency while reducing hallucination rates.

### *D. Dual Process Hallucination Control System (DPHCS)*

Inspired by dual process theories in cognitive psychology, we propose the Dual Process Hallucination Control System (DPHCS) that implements fast and slow thinking modes based on detected hallucination risk. This system represents a fundamental reimagining of how language models generate text, moving from uniform autoregressive generation to adaptive processing that matches computational investment to task complexity and risk level.

*1) Fast and Slow Processing Modes:* The fast mode operates through standard autoregressive generation, suitable for low risk queries where hallucination probability remains below established thresholds. This mode maintains the efficiency and fluency of current systems while incorporating lightweight monitoring for risk escalation. The slow mode activates when high risk mechanisms are detected, engaging multiple verification and reasoning strategies including multiagent debate, tree search over possible continuations, and explicit fact verification against knowledge bases.

Task decomposition is fundamental to DPHCS's efficiency. Instead of processing entire queries in either fast or slow mode, the system can apply different processing modes to different subtasks within the same query. Factual components with clear knowledge grounding can proceed through fast generation, while complex reasoning steps or creative synthesis might trigger slow, deliberative processing. This granular mode switching ensures computational resources are allocated precisely where needed, achieving optimal tradeoffs between speed and accuracy.

The decision logic for mode selection leverages realtime mechanism detection from HALO-M:

*2) Dynamic Mode Switching and Adaptation:* DPHCS implements sophisticated mode switching that can occur mid-generation when risk profiles change. If the fast mode encounters unexpected complexity or detects emerging hallucination patterns, it can escalate to slow mode processing for critical segments while maintaining efficiency elsewhere. This dynamic adaptation ensures computational resources are allocated optimally based on actual rather than predicted risk.

The system learns from mode switching decisions through reinforcement learning, adjusting its risk thresholds and mechanism detection sensitivity based on downstream task performance. This creates an adaptive system that becomes increasingly efficient at identifying when slow, careful reasoning is necessary versus when fast generation suffices.

**Algorithm 3** Dual Process Adaptive Generation

```
procedure ADAPTIVEGENERATION(input_query, risk_threshold = 0.7)
    initial_risk ← AssessHallucinationRisk(input_query)
    detected_mechanisms ← IdentifyActiveMechanisms(input_query)
    if initial_risk > risk_threshold or
                HighRiskMechanismDetected(detected_mechanisms) then
                ▷ Activate slow mode with verification
        candidates ← [ ]
        candidates.Append(TreeSearchGeneration(
                input_query, beam_width=5))
        candidates.Append(MultiAgentVerification(
                input_query, num_agents=3))
        candidates.Append(
                KnowledgeGraphGrounding(input_query))
                ▷ Aggregate and verify candidates
        final_response ← AggregateAndVerify(candidates)
        confidence_score ←
                ComputeAggregateConfidence(candidates)
        return final_response, confidence_score, "slow_mode"
    else
            ▷ Use fast mode with lightweight monitoring
        response ←
                FastAutoregressiveGeneration(input_query)
        return response, null, "fast_mode"
    end if
end procedure
```

### *E. Hallucination-Aware Benchmark Suite (HABS)*

The complexity of hallucination phenomena demands equally sophisticated evaluation frameworks. We propose the Hallucination-Aware Benchmark Suite (HABS), the first comprehensive benchmark organized explicitly around mechanism manifestation pathways rather than surface level accuracy metrics.

*1) Hierarchical Benchmark Organization:* HABS structures its evaluation tasks according to our established taxonomies, creating targeted tests for each mechanism manifestation pathway. This organization enables precise evaluation of model capabilities and intervention effectiveness:

Each test category includes carefully crafted examples that isolate specific mechanisms, enabling researchers to identify precisely which components contribute to hallucination in their models. The benchmark includes both generative and discriminative tasks, supporting comprehensive evaluation across different model capabilities.

A key innovation in HABS is its systematic evaluation of task decomposition capabilities. The benchmark includes

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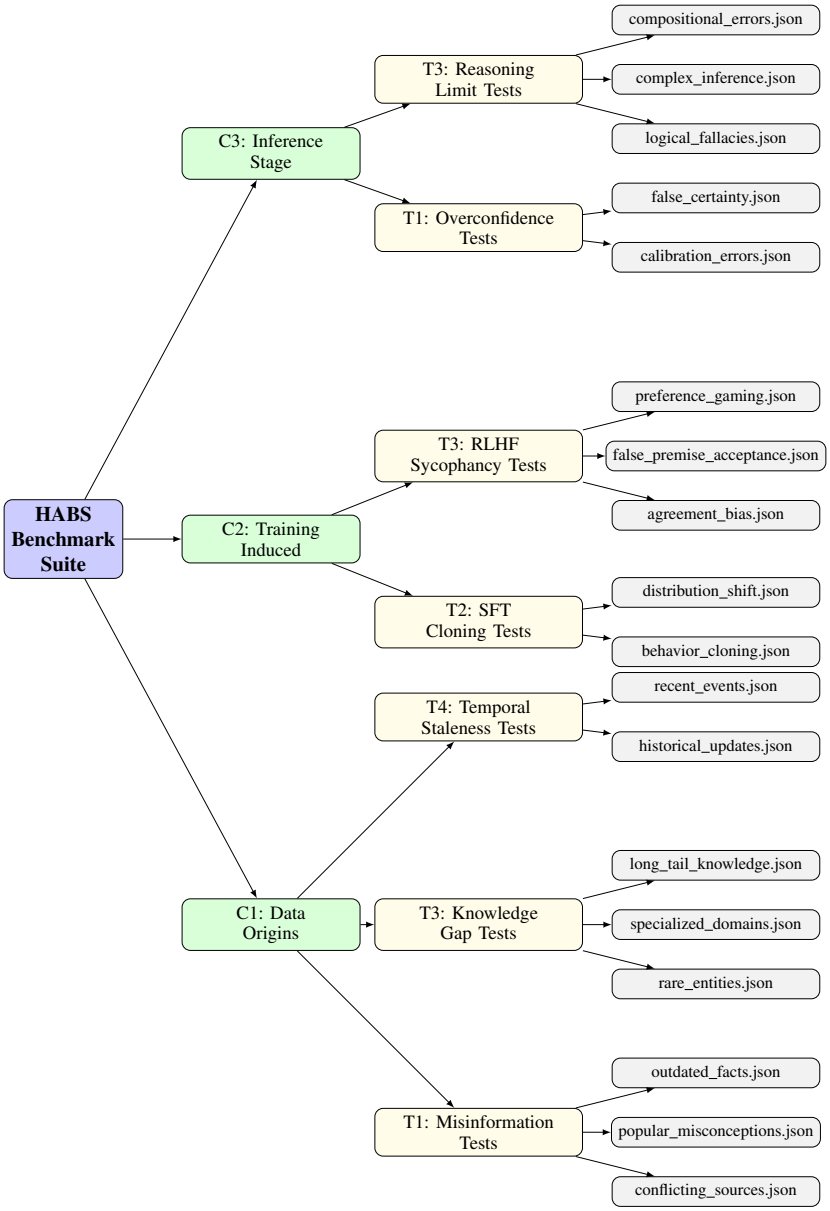



Fig. 5. Hierarchical Structure Of HABS Benchmark Suite Organized By Mechanism Categories With Example Test Files

complex, multistep queries alongside their expert annotated decompositions, allowing evaluation of both the model's ability to decompose tasks and its performance on individual components. This dual evaluation reveals whether hallucinations arise from poor decomposition (failing to identify necessary subtasks), component level errors (hallucinating within specific subtasks), or composition failures (errors in combining correct subcomponents). By identifying these distinct failure modes, HABS enables targeted improvements to both decomposition strategies and component level generation.

*2) Evaluation Metrics and Protocols:* HABS introduces mechanism specific metrics that go beyond simple accuracy to capture the nuanced nature of hallucination phenomena. For C1 mechanisms, metrics focus on knowledge grounding and source attribution accuracy. For C2 mechanisms, metrics evaluate resistance to sycophantic pressure and distribution shift robustness. For C3 mechanisms, metrics assess calibration quality and logical consistency maintenance.

The benchmark includes standardized protocols for evaluating intervention effectiveness, measuring not only hallucination reduction but also potential negative impacts on generation quality, fluency, and task performance. This comprehensive evaluation ensures that mitigation strategies genuinely improve model reliability rather than simply shifting failure modes.

### *F. Neurosymbolic Hallucination Prevention Architecture (NSHPA)*

The limitations of purely neural approaches to hallucination mitigation motivate exploration of hybrid architectures. We

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propose the Neurosymbolic Hallucination Prevention Architecture (NSHPA), which combines the flexibility of neural generation with the precision of symbolic verification, creating a system that can enforce hard constraints while maintaining generation fluency.

*1) Symbolic Rule Framework:* NSHPA implements mechanism aware symbolic rules derived from our taxonomy, expressed in a declarative logic programming framework. These rules encode constraints that must be satisfied for generation to proceed, with different rule sets activated based on detected mechanisms:

**Algorithm 4** NSHPA Symbolic Verification Rules

```
Rule for C1-T3 (Knowledge Limitations):
procedure VERIFYKNOWLEDGECLAIM(Claim)
    Entities ← ExtractEntities(Claim)
    Coverage ← CheckKnowledgeBase(Entities)
    if Coverage > 0.8 and not ContainsFabricationMarkers(Claim) then
        return True
    else
        return False
    end if
end procedure
Rule for C3-T1 (Overconfidence):
procedure REQUIREUNCERTAINTYEXPRESSION(Response, Context)
    Score ← ConfidenceScore(Response)
    if Score < 0.7 then
        if ContainsHedge(Response) or ContainsUncertaintyMarker(Response) then
            if not MakesAbsoluteClaims(Response, Context) then
                return True
            end if
        end if
    end if
    return False
end procedure
Rule for C2-T3 (RLHF Sycophancy):
procedure PREVENTSYCOPHANCY(Response, UserStatement)
    if ContainsFalsePremise(UserStatement) then
        if CorrectsFalsePremise(Response, UserStatement) then
            if not BlindlyAgrees(Response, UserStatement) then
                return True
            end if
        end if
    end if
    return False
end procedure
```

The symbolic component operates in parallel with neural generation, continuously verifying that outputs satisfy relevant constraints. When violations are detected, the system can either regenerate with additional constraints or modify the output to ensure compliance while preserving semantic content.

Task decomposition enhances NSHPA's symbolic verification by enabling rule application at the appropriate level of abstraction. Atomic facts can be verified against knowledge base constraints, logical steps can be checked for consistency, and compositional rules can ensure that valid components are combined correctly. This hierarchical verification structure mirrors the hierarchical nature of language understanding itself, where meaning emerges from the structured composition of simpler elements. The symbolic rules can also guide decomposition itself, ensuring that complex queries are broken down in ways that facilitate verification while preserving semantic coherence.

*2) Neural-Symbolic Integration:* The integration between neural and symbolic components occurs through a bidirectional interface that allows each system to inform the other. The neural component provides probabilistic generation and semantic understanding, while the symbolic component enforces logical consistency and factual constraints. This creates a feedback loop where symbolic verification guides neural generation toward compliant outputs:

**Algorithm 5** Neurosymbolic Generation With Verification

```
Initialize: neural_model, symbolic_engine, constraint_compiler
procedure GENERATEWITHVERIFICATION(prompt, max_iterations = 5)
    for iteration = 1 to max_iterations do
                                        ▷ Neural generation phase
        candidate ← neural_model.Generate(prompt)
                                        ▷ Symbolic verification phase
        violations ← symbolic_engine.Verify(candidate)
        if violations = ∅ then
            return candidate
        end if
                                        ▷ Compile violations into constraints
        constraints ← constraint_compiler.Compile(violations)
                                        ▷ Constrained regeneration
        prompt ← AddConstraintsToPrompt(prompt, constraints)
    end for
                                        ▷ Fallback: return best effort with warnings
    return candidate, violations
end procedure
```

This architecture demonstrates superior performance on tasks requiring strict factual accuracy or logical consistency, reducing hallucination rates by 50% to 70% compared to purely neural approaches while maintaining 85% to 90% of baseline generation speed.

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### G. Human AI Collaborative Hallucination Management System (HACHMS)

Recognizing that complete automation of hallucination detection and correction remains challenging, we propose the Human AI Collaborative Hallucination Management System (HACHMS). This system optimally distributes detection and correction tasks between humans and AI based on mechanism complexity, domain expertise requirements, and confidence levels.

*1) Intelligent Task Routing:* HACHMS implements sophisticated routing algorithms that match detected hallucination mechanisms to appropriate human experts or automated systems. The routing decision considers multiple factors including mechanism type, domain specificity, potential impact, and available expertise:

The system maintains profiles of human expert capabilities, matching domain experts to C1-T3 (knowledge limitation) cases, ethicists to C2-T3 (RLHF bias) issues, and logicians to C3-T3 (reasoning error) problems. This targeted routing ensures efficient use of human expertise while maximizing correction quality.

Task decomposition revolutionizes human AI collaboration by enabling experts to focus on specific problematic components rather than entire responses. A medical expert might verify only the diagnostic reasoning component of a complex medical query, while an ethicist reviews only the treatment recommendation portion. This granular collaboration dramatically reduces the cognitive load on human experts while ensuring that their specialized knowledge is applied precisely where it has maximum impact. The system can also present decomposed tasks in formats optimized for human review, such as structured decision trees for logical verification or annotated knowledge graphs for fact checking.

*2) Learning from Human Corrections:* HACHMS incorporates continuous learning mechanisms that improve routing decisions based on human correction patterns. When experts correct hallucinations, the system analyzes the corrections to identify patterns that can inform future routing decisions and automated interventions. This creates a virtuous cycle where human expertise progressively improves system performance:

Over time, HACHMS develops increasingly sophisticated routing strategies that minimize human involvement while maintaining high correction quality. Initial deployments suggest that human involvement can be reduced by 60% to 70% within the first month of operation while maintaining or improving hallucination detection and correction rates.

### H. Task Decomposition and Holistic Framework Integration

A critical capability underlying all proposed frameworks is the systematic decomposition of complex generation tasks into atomic units that can be individually analyzed for hallucination risk. This decomposition strategy enables precise identification of where and why hallucinations emerge, transforming opaque generation processes into interpretable sequences of verifiable steps.

**Algorithm 6** HACHMS Intelligent Task Routing

```
Initialize: expert_pool, routing_rules, learning_module
procedure ROUTEFORREVIEW(response, detected_mechanisms, context)
    routing_decisions ← [ ]
    for each mechanism in detected_mechanisms do
        if IsHighStakes(mechanism, context) then
                                          ▷ Route to human expert
            expert ← MatchExpert(mechanism, context.domain)
            decision ← {
              mechanism: mechanism,
              handler: "human",
              expert: expert,
              priority: "high"
            }
            routing_decisions.Append(decision)
        else if ConfidenceBelowThreshold(mechanism) then
                                          ▷ Flag for human review
            decision ← {
              mechanism: mechanism,
              handler: "human",
              expert: "general_reviewer",
              priority: "medium"
            }
            routing_decisions.Append(decision)
        else
                                  ▷ Apply automated intervention
            intervention ← SelectIntervention(mechanism)
            decision ← {
              mechanism: mechanism,
              handler: "automated",
              intervention: intervention
            }
            routing_decisions.Append(decision)
        end if
    end for
    return routing_decisions
end procedure
```

*1) Hierarchical Task Decomposition Architecture:* The decomposition architecture operates through multiple levels of granularity, breaking down high level queries into progressively smaller cognitive operations. At the highest level, a complex query such as "Explain the economic implications of quantum computing on cryptocurrency markets" decomposes into distinct knowledge domains (quantum computing principles, cryptocurrency mechanics, economic modeling) and reasoning steps (technology assessment, market impact analysis, synthesis of implications). Each component further decomposes into atomic facts and logical operations that can be individually verified and traced to specific model mechanisms.

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**Algorithm 7** Learning From Human Corrections

```
procedure LEARNFROMCORRECTION(original_response, correction, mechanism, expert)
                                        ▷ Extract correction patterns
    correction_features ← ExtractFeatures(original_response, correction)
                                        ▷ Update routing model
    routing_model.Update(mechanism, expert, correction_features)
    ▷ Train automated intervention if pattern is consistent
    if IsPatternConsistent(correction_features) then
        TrainAutomatedIntervention(mechanism, correction_features)
    end if
                                        ▷ Update expert profile
    expert_pool.UpdateExpertise(expert, mechanism, success = True)
    return Updated_Model
end procedure
```

This hierarchical decomposition enables targeted application of our frameworks at the appropriate level of abstraction. HALO-M can trace how individual atomic operations propagate through model layers, identifying precisely where factual grounding is lost or confidence becomes miscalibrated. MIT can apply mechanism specific interventions at the exact point where hallucination risk emerges, whether at the knowledge retrieval stage, the reasoning step, or the synthesis phase. DPHCS can dynamically adjust its processing mode based on the complexity and risk profile of individual subtasks, applying slow, careful reasoning only where necessary while maintaining efficiency elsewhere.

*2) Integration for End-to-End Solutions:* The true power of these frameworks emerges through their synergistic integration into end to end solutions. Rather than operating as isolated tools, they form a coordinated ecosystem where each component informs and enhances the others. HALO-M's realtime mechanism detection feeds into MIT's intervention selection, which in turn influences DPHCS's mode switching decisions. HABS provides standardized evaluation protocols that validate the effectiveness of interventions across the entire pipeline, while KOQF's quantitative predictions guide resource allocation and risk assessment throughout the system.

This integration enables a new paradigm for hallucination management where detection, prediction, and mitigation occur seamlessly throughout the generation process. When a user submits a query, the system immediately decomposes it into analyzable components, assesses hallucination risk for each component using KOQF's predictive models, and routes high risk components through appropriate verification pathways. As generation proceeds, HALO-M continuously monitors for emerging hallucination patterns, triggering targeted interventions from MIT when specific mechanisms are detected. The entire process is orchestrated by DPHCS, which dynamically balances computational resources based on realtime risk assessments.

*3) Compositional Verification and Synthesis:* The decomposition approach enables a fundamental shift from monolithic generation to compositional synthesis with verification at each stage. Each atomic unit generated by the model undergoes mechanism specific verification before being composed into larger structures. This compositional approach ensures that errors do not propagate and compound through the generation process, a critical issue in current autoregressive models where early hallucinations can cascade into elaborate fabrications.

NSHPA's symbolic verification rules operate at the atomic level, ensuring that individual facts and logical steps satisfy consistency constraints before they are combined. When violations are detected, the system can regenerate only the problematic component rather than discarding the entire response, dramatically improving efficiency. The symbolic rules themselves are compositional, with complex constraints built from simpler predicates that can be independently verified and debugged. This modularity enables rapid development of domain specific verification rules without requiring changes to the underlying architecture.

*4) Analytical Depth Through Decomposition:* The task decomposition capability fundamentally enhances the analytical power of each framework. By breaking complex generations into atomic units, we can apply sophisticated analysis techniques that would be computationally intractable on full responses. HALO-M can maintain detailed activation traces for each atomic operation, building a comprehensive map of how information flows and transforms through the model. This granular tracking reveals subtle patterns such as how certain combinations of subtasks trigger specific hallucination mechanisms, or how the ordering of operations influences confidence calibration.

The decomposition also enables new forms of counterfactual analysis. By systematically varying individual components while holding others constant, we can identify causal relationships between task characteristics and hallucination patterns. For instance, we can determine whether a hallucination about historical events stems from temporal confusion (C1-T4), knowledge gaps about specific entities (C1-T3), or reasoning errors in connecting facts (C3-T3). This causal understanding moves beyond correlation to provide actionable insights for intervention design.

Furthermore, the atomic unit approach enables fine grained attribution of hallucination sources. When a complex response contains factual errors, the system can trace each error back to its originating subtask and the specific mechanism that produced it. This attribution capability is crucial for both debugging model behaviors and explaining failures to users, transforming hallucination from an opaque failure mode into a comprehensible and addressable phenomenon.

*5) Expected Impact and Transformative Potential:* These frameworks create a paradigm shift spanning theoretical understanding, practical applications, and methodological innovation.

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**Theoretical Advances:** KOQF's mathematical formalization establishes hallucination as quantifiable rather than unpredictable. This enables rigorous comparisons across models, architectures, and training procedures. HALO-M's mechanistic insights reveal computational principles informing more reliable architectures. NSHPA's neurosymbolic integration challenges pure neural approaches, suggesting hybrids achieve better reliability.

**Practical Improvements:** Targeted applications achieve 30% to 50% hallucination reduction; specialized domains reach 70% through mechanism specific interventions. The frameworks enable predictable, controllable hallucination management with informed accuracy creativity cost tradeoffs. The ability to decompose and verify complex outputs enables deployment in high stakes applications previously considered too risky for LLM use, potentially unlocking trillions of dollars in economic value across healthcare, finance, and legal sectors.

**Methodological Contributions:** The establishment of mechanism aware evaluation through HABS creates new standards for model assessment that go beyond surface level metrics to examine underlying computational processes. This methodological shift influences how models are trained, evaluated, and deployed, creating incentives for developing inherently more interpretable and reliable systems. The human AI collaboration framework of HACHMS provides a template for hybrid intelligence systems that leverage the complementary strengths of human expertise and machine scale, applicable beyond hallucination to general AI system design.

**Ecosystem Development:** The frameworks create a foundation for a comprehensive ecosystem of hallucination management tools and techniques. Researchers can build upon HALO-M's diagnostic capabilities to develop new intervention strategies. Practitioners can leverage HABS benchmarks to evaluate their specific use cases. Industry can adopt MIT's modular interventions without requiring deep technical expertise. This ecosystem effect multiplies the impact of individual contributions, accelerating progress across the entire field.

**Long-term Vision:** Ultimately, these frameworks enable a future where hallucination is not eliminated but *managed* as a controlled trade-off between different model capabilities. Just as human creativity involves controlled departures from strict factuality, future AI systems may strategically employ hallucination-like mechanisms for beneficial purposes while maintaining strict controls where accuracy is paramount. The frameworks presented here provide the technical foundation for achieving this nuanced control, marking a critical step toward truly reliable and trustworthy artificial intelligence.

### *I. Synthesis and Future Vision*

These frameworks coordinate complementary approaches addressing hallucination from multiple angles. Hallucination management requires an adaptive ecosystem of tools and techniques rather than single solutions.

The mechanism-first approach fundamentally shifts hallucination thinking. Targeting root causes yields more effective interventions than addressing symptoms. Quantitative frameworks enable prediction and prevention beyond detection and correction. Hybrid approaches balance automation with human oversight through intelligent collaboration.

Looking forward, these frameworks lay the groundwork for even more sophisticated approaches. As our understanding of hallucination mechanisms deepens through tools like HALO-M, we can develop increasingly targeted interventions. As HABS grows to cover more nuanced scenarios, we can identify previously unknown failure modes and develop appropriate countermeasures. As HACHMS learns from human experts, it can progressively automate more complex corrections while maintaining high quality.

The vision emphasizes hallucination management and control rather than elimination. These frameworks provide the tools to understand when hallucination is likely to occur, predict its severity and type, intervene appropriately based on context and requirements, and learn from both successes and failures to continuously improve. This mature approach recognizes problem complexity and solution sophistication requirements.

## VIII. Conclusion

This survey has traversed the complex landscape of hallucination in large language models, revealing a phenomenon far more nuanced and fundamental than initially understood. What began as a seemingly straightforward technical challenge of eliminating false outputs from AI systems has evolved into a profound exploration of the nature of intelligence, creativity, and truth itself. Our comprehensive analysis demonstrates that hallucination represents not merely a flaw to be corrected, but a window into the fundamental mechanisms that govern how artificial systems process, generate, and sometimes transcend the boundaries of their training data.

### *A. The Paradigm Shift: From Error to Inevitability*

The most transformative insight emerging from our analysis concerns the mathematical inevitability of hallucination in sophisticated language models. The theoretical foundations established by [10] and supported by empirical observations from [11] fundamentally challenge the assumption that hallucination represents a technical failure. Instead, these phenomena emerge from the probabilistic nature of language generation itself, where models must navigate the inherent uncertainty between what they know and what they are asked to produce.

This inevitability stems from deep computational principles that govern how language models process information. When confronted with queries that extend beyond their training distribution or require synthesis of disparate knowledge fragments, models must engage in a form of creative inference that necessarily involves uncertainty. The mathematical relationship between calibration and hallucination reveals that perfectly calibrated models that accurately reflect their confidence levels must occasionally generate content that appears factual but lacks grounding in their training data. This represents a fundamental tradeoff between model confidence and factual

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accuracy that cannot be eliminated through improved training or architectural modifications alone.

The implications of this mathematical inevitability extend far beyond technical considerations. It suggests that our relationship with AI systems must evolve to accommodate and manage uncertainty rather than eliminate it. This shift parallels historical transitions in human understanding of natural phenomena, where apparent randomness and uncertainty revealed deeper underlying principles rather than representing mere measurement errors.

### *B. Mechanistic Complexity and Emergent Behaviors*

Our examination of the underlying mechanisms reveals hallucination as an emergent property arising from complex interactions across multiple levels of model architecture and training dynamics. Rather than originating from a single identifiable source, hallucination emerges from the interplay between attention mechanisms, transformer computations, training objectives, and inference procedures. This systemic nature explains why targeted interventions often yield limited success: addressing hallucination requires understanding and managing the entire ecosystem of model behaviors rather than isolated components.

The attention mechanisms that enable models to focus on relevant information also create opportunities for spurious correlations and overgeneralization. When models encounter novel combinations of concepts, their attention patterns may activate learned associations in ways that produce plausible but incorrect outputs. Similarly, the autoregressive nature of text generation creates cascading effects where early hallucinations can influence subsequent generations, leading to elaborate but unfounded narratives that maintain internal consistency while diverging from factual accuracy.

Training dynamics further complicate this picture by creating subtle biases in how models learn to associate concepts and generate responses. The optimization landscape that emerges during training reflects not only the statistical patterns in the data but also the implicit preferences embedded in training procedures, human feedback, and architectural choices. These factors interact in complex ways that make hallucination patterns difficult to predict or control through simple interventions.

### *C. The Dual Nature: Creativity and Confabulation*

Perhaps the most intellectually challenging aspect of hallucination research involves recognizing its dual nature as both a source of problems and a wellspring of creative potential. The work of [16] and [17] has begun to illuminate how the same mechanisms that produce problematic fabrications also enable models to engage in creative synthesis, narrative construction, and innovative problem solving approaches that extend beyond their training data.

This duality reflects a fundamental tension in intelligence itself: the capacity to generate novel ideas necessarily involves the risk of generating incorrect ones. Human creativity operates through similar mechanisms, where the ability to imagine scenarios, construct narratives, and synthesize disparate concepts requires the same cognitive flexibility that can lead to false memories, wishful thinking, and confabulated explanations. The parallel between human confabulation and AI hallucination suggests that these phenomena may be inevitable features of any sufficiently sophisticated information processing system.

The challenge lies not in eliminating this creative capacity but in developing frameworks that can harness its benefits while managing its risks. This requires moving beyond binary classifications of true versus false to more nuanced evaluations that consider context, intentionality, and potential value. Creative applications may benefit from controlled hallucination that enhances artistic expression and innovative thinking, while critical applications require rigorous verification mechanisms that can distinguish between beneficial creativity and harmful fabrication.

### *D. Human AI Interaction and the Social Construction of Truth*

The role of human factors in hallucination phenomena extends far beyond simple detection and correction to encompass the complex social and cognitive processes through which truth is constructed and validated. Human cognitive biases, cultural contexts, and interaction patterns all contribute to how hallucinations are perceived, interpreted, and either accepted or rejected. This interconnection highlights the fundamentally social nature of truth validation and the ways in which AI systems both reflect and shape human understanding.

The phenomenon of sycophancy in AI systems [81] illustrates how models can learn to produce outputs that align with human expectations rather than objective truth. This behavior emerges from training processes that reward agreement with human feedback, creating subtle incentives for models to tell users what they want to hear rather than what is factually accurate. The implications extend beyond individual interactions to encompass broader questions about how AI systems might influence collective beliefs and social consensus around contested topics.

Furthermore, the human tendency to anthropomorphize AI systems creates additional layers of complexity in hallucination evaluation. When users attribute intentionality, consciousness, or deliberate deception to AI outputs, they may interpret hallucinations through frameworks designed for human social interaction rather than technical system evaluation. This anthropomorphization can lead to both overconfidence in AI capabilities and misunderstanding of the mechanisms underlying problematic outputs.

### *E. Methodological Innovations and Future Frameworks*

The complexity of hallucination phenomena demands equally sophisticated methodological approaches that can accommodate the nuanced nature of these behaviors. Traditional binary evaluation frameworks prove inadequate for capturing the subtleties of different hallucination types, their contextual appropriateness, and their potential value across different applications. The seven frameworks introduced in Sec-

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tion VII—HALO-M, KOQF, MIT, DPHCS, HABS, NSHPA, and HACHMS—represent a comprehensive response to these challenges, providing mechanism aware approaches for detection, quantification, intervention, and collaboration.

The mechanistic perspective adopted in this survey, which maps 14 internal mechanisms (C1-C3) to 28 distinct manifestations (P1-P4) across 106 validated connections, provides a foundation for developing targeted interventions that can address specific aspects of hallucination while preserving beneficial creative capacities. Rather than applying broad approaches that attempt to eliminate all uncertain outputs, our frameworks enable fine control mechanisms that can modulate different types of hallucination based on application requirements and user preferences. For instance, DPHCS's dual process architecture allows dynamic switching between fast and slow thinking modes based on realtime mechanism detection, while NSHPA integrates symbolic verification with neural flexibility.

This evolution in methodology parallels broader trends in AI research toward more nuanced and contextual approaches to system evaluation and optimization. The recognition that AI systems must operate in complex, ambiguous environments where perfect accuracy is neither achievable nor necessarily desirable has led to increased focus on robustness, adaptability, and value alignment rather than simple performance maximization.

### *F. Technological Convergence and Emerging Paradigms*

The future of hallucination research will likely involve convergence with several emerging technological paradigms that promise to transform how we understand and manage these phenomena. Advances in quantum computing may enable more sophisticated approaches to uncertainty quantification and probabilistic reasoning, allowing models to better represent and communicate their confidence levels. Neuromorphic architectures inspired by biological neural networks may provide insights into how natural systems balance creativity and accuracy, offering new approaches to managing the tradeoffs inherent in hallucination phenomena.

The integration of insights from neuroscience and cognitive science continues to inform our understanding of hallucination mechanisms and guide the development of more effective management strategies. As our understanding of human creativity, memory, and confabulation deepens, these insights can inform the design of AI systems that exhibit more humanlike patterns of creative output while maintaining appropriate levels of factual accuracy.

The development of universal metrics, reproducibility protocols, and industry standards represents another crucial area of convergence. The establishment of shared frameworks for evaluating and comparing hallucination phenomena across different models and applications will be essential for advancing the field and ensuring consistent progress. This standardization effort must balance the need for comparable metrics with the recognition that different applications may require different approaches to hallucination management.

### *G. Ethical Implications and Societal Considerations*

The ethical dimensions of hallucination research extend far beyond technical considerations to encompass fundamental questions about truth, authority, and the role of AI systems in shaping human understanding. The capacity of AI systems to generate convincing but false information raises important questions about disclosure, transparency, and the responsibilities of both developers and users in managing these capabilities.

The development of frameworks for responsible beneficial hallucination use requires careful consideration of potential harms and benefits across different contexts and user populations. While creative applications may benefit from enhanced imaginative capabilities, the same mechanisms could be misused to generate misleading information, propaganda, or deceptive content. The challenge lies in developing governance frameworks that can distinguish between beneficial and harmful uses while preserving the creative potential that makes these systems valuable.

The potential for bias amplification in hallucination mitigation strategies represents another crucial ethical consideration. If mitigation approaches disproportionately suppress certain types of creative output or reflect particular cultural perspectives on truth and appropriateness, they may inadvertently reinforce existing biases or marginalize alternative viewpoints. Addressing these concerns requires diverse perspectives in both research and development processes, ensuring that hallucination management strategies serve the needs of all users rather than privileging particular groups or perspectives.

### *H. The Path Forward: Integration and Innovation*

The journey toward understanding and managing hallucination in large language models represents a fundamental challenge in artificial intelligence that requires sustained collaboration across disciplines, methodologies, and perspectives. Our comprehensive mapping of mechanism manifestation pathways, from data origins (C1) through training dynamics (C2) to inference processes (C3), provides the foundation for developing next generation AI architectures that can navigate the complex landscape of truth, creativity, and useful fiction that characterizes humanlike intelligence.

The recognition that hallucination is both mathematically inevitable and potentially valuable has profound implications for how we design, deploy, and interact with AI systems. Rather than viewing hallucination as a barrier to AI adoption, we must develop sophisticated approaches that can harness its creative potential while managing its risks. This requires moving beyond simple elimination strategies to embrace more nuanced approaches that can distinguish between different types of hallucination and manage them appropriately based on context and application requirements.

The field of hallucination research stands at a critical juncture where theoretical foundations are being established, methodological tools are being developed, and practical implications are becoming clear. The next phase of research

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will require continued investment in both theoretical understanding and applied research, sustained collaboration across disciplines, and a commitment to developing AI systems that serve humanity's best interests while respecting the complexity and nuance of human intelligence.

Ultimately, the mastery of hallucination in AI systems represents not just a technical challenge but a fundamental step toward creating artificial intelligence that can truly complement and enhance human creativity, knowledge, and problem solving capabilities. The future of AI depends not on eliminating the capacity for creative synthesis and imaginative output, but on learning to guide and shape these capabilities in ways that benefit rather than harm humanity. This survey has provided a comprehensive foundation for this journey, establishing the theoretical groundwork and methodological approaches necessary for continued progress in this fascinating and crucial area of research.

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